A Simulation Framework for Detecting Causal Microbial Taxa in Built-Environment Allergy Studies:

Power Analysis for a Household Sensor Network

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Abstract: The built-environment microbiome represents a largely unexplored determinant of human respiratory and allergic health. Here we present a simulation framework calibrated to empirical temporal dynamics from a 61-day daily sampling study of residential sink P-trap communities (Hill et al., 2026). Our framework simulates 100 microbial taxa with AR(1) dynamics (CV range 5-25%), five environmental covariates, and daily allergy scores driven by 5 causal taxa with effect sizes of 2-10%. We evaluate four analytical approaches: elastic net, per-taxon univariate models with BH FDR correction, mixed-effects models, and mediation analysis that decomposes taxon-allergy associations into direct vs environment-confounded paths. Elastic net achieves >93% sensitivity with 20+ households but suffers high FDR (>92%). Univariate models reveal a structural problem: environment-linked non-causal taxa generate 10-15 persistent false positives because environmental parameters simultaneously drive both taxon abundances and allergy symptoms. Increasing sample size improves sensitivity but does not eliminate these confounded associations. Mediation analysis reduces false positives from ~20 to ~9 by correctly classifying environment-confounded taxa, but residual false positives remain and the observed FDR (~70%) still exceeds the nominal 5% threshold. These results demonstrate that environmental confounding--not simply multiple testing--is the primary driver of false discoveries in built-environment microbiome studies.

Keywords: built-environment microbiome, allergy, simulation, power analysis, elastic net, longitudinal study design, sensor network

Introduction

1. INTRODUCTION

The built environment—comprising the indoor spaces where people spend approximately 90% of their time—harbors complex microbial communities that residents inhale, ingest, and contact daily (Adams et al., 2015). Epidemiological studies have linked indoor microbial exposures to both protective and pathogenic effects on respiratory health: farm-dwelling children exposed to diverse environmental microbiota show reduced asthma prevalence (Ege et al., 2011), while indoor mold and fungal exposures are consistently associated with allergic sensitization and exacerbation (Mendell et al., 2011). Despite these associations, the specific microbial taxa driving allergic responses in typical residential settings remain largely unidentified, in part because most studies rely on cross-sectional sampling that cannot capture the dynamic interplay between fluctuating microbial exposures and symptom trajectories.

Recent technological advances are converging to make continuous household microbiome monitoring feasible. Environmental DNA sequencing costs continue to fall, miniaturized particulate and volatile organic compound sensors enable continuous air quality monitoring, and mobile health platforms allow real-time symptom tracking. We envision a household sensor network (Figure 1) comprising microbial sampling units deployed in key indoor locations (kitchen, bathroom, HVAC system) alongside environmental monitors for temperature, humidity, particulate matter (PM2.5), carbon dioxide (CO2), and fungal spore concentration, coupled with a mobile application through which household members record daily allergy severity scores.

A critical question for the design of such studies is: given realistic levels of microbial community variability, environmental noise, and between-person heterogeneity, how many households and how many days of sampling are required to detect microbial taxa that contribute modestly (2-10%) to allergy symptom severity?

Answering this question requires a realistic simulation framework that captures the key statistical challenges of the problem: high-dimensional predictors (100+ taxa), compositional constraints, temporal autocorrelation in both exposures and outcomes, hierarchical structure (observations nested within persons nested within households), and the need for variable selection in the presence of many null predictors.

In this paper, we present such a framework. We calibrate the day-to-day dynamics of simulated microbial taxa to empirical data from Hill et al. (2026), who performed the first daily-resolution longitudinal characterization of residential sink P-trap bacterial communities over 61 consecutive days. Their study revealed that even adjacent sinks in a shared bathroom can exhibit dramatically different temporal stability profiles—from highly deterministic communities with a coefficient of variation (CV) of 4.9% and strong temporal autocorrelation, to stochastic communities with a CV of 26.5% and no temporal predictability. This range provides an empirical basis for simulating the diversity of temporal behaviors expected across taxa in a household microbiome study.

We simulate longitudinal data for varying numbers of households, each containing multiple family members who provide daily allergy reports. A sparse set of 5 "causal" taxa, embedded among 95 null taxa, drive allergy severity with effect sizes ranging from 2% to 10%. We then apply elastic net regularized regression and evaluate the power to correctly identify the causal taxa as a function of study size, duration, and effect magnitude.

Figure 1: Household Sensor Network

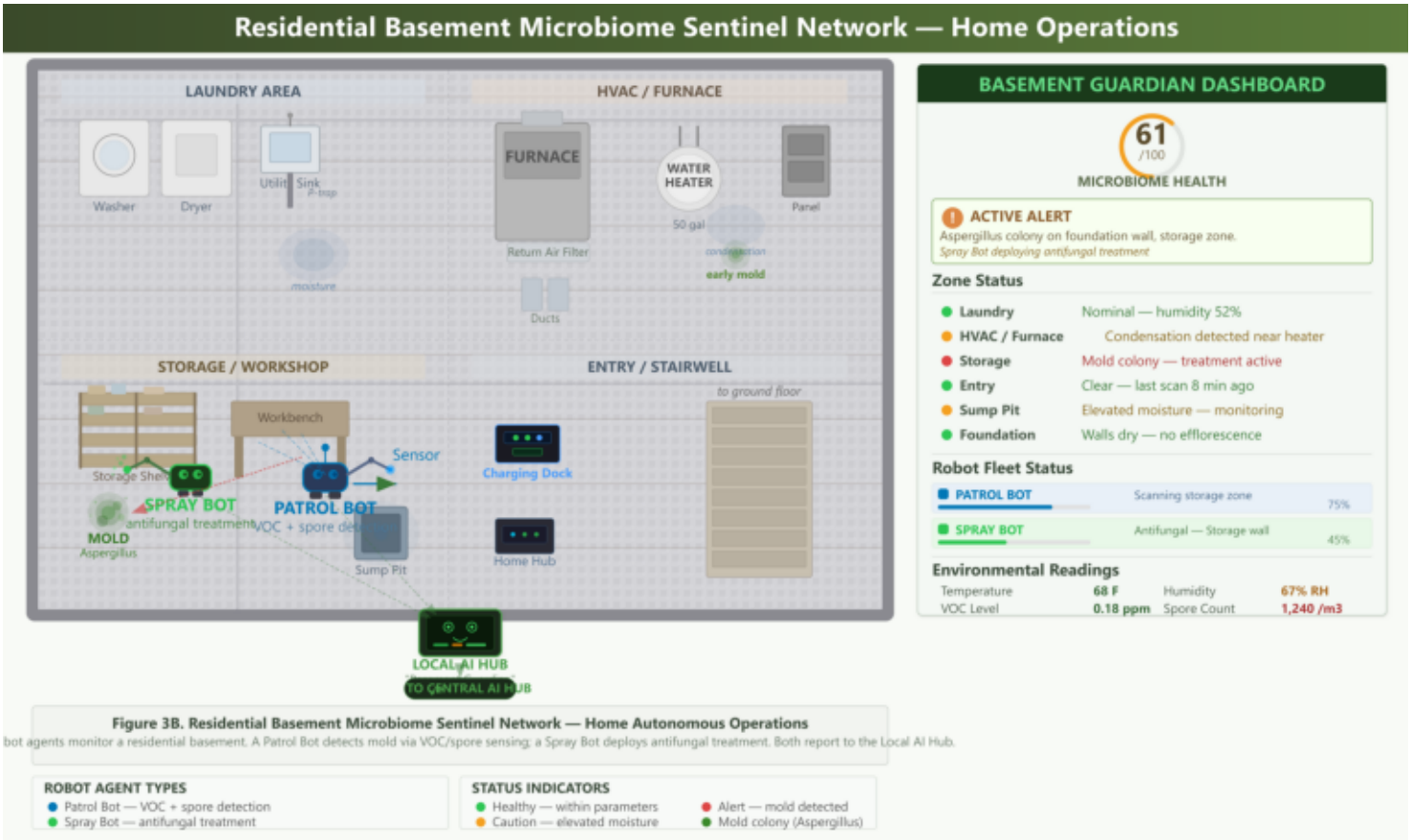


Figure 1. Conceptual schematic of the household sensor network for integrated microbiome-environment-health monitoring. Microbial sampling units collect daily environmental specimens for 16S rRNA sequencing. Environmental sensors continuously monitor temperature, humidity, PM2.5, CO2, and fungal spore concentrations. Household members report daily allergy severity via a mobile application.

Methods

2. METHODS

2.1 Calibration Data: Hill et al. (2026)

We calibrated the temporal dynamics of our simulated microbial communities to the empirical observations of Hill et al. (2026), who conducted a 61-day daily sampling study of bacterial communities in paired residential bathroom sink P-traps. This study represents the highest temporal resolution characterization of residential microbial communities published to date.

Hill et al. sequenced full-length 16S rRNA genes from daily swab samples of two sinks (A and B) sharing identical plumbing infrastructure, environmental conditions, and cleaning regimes but differing in primary user and usage patterns. Their findings revealed two distinct dynamical regimes:

> Stable regime (Sink A): Community composition exhibited a coefficient of variation of 4.9%, significant temporal autocorrelation (Mantel test, $p = 0.001$), and deterministic dynamics in which time explained 49.9% of community variation. This sink was primarily used for handwashing, toothbrushing, and shaving. The community was enriched in aerobic, skin-associated taxa.
>
> Volatile regime (Sink B): Community composition showed a CV of 26.5%, no significant temporal autocorrelation ($p = 0.53$), and stochastic dynamics with minimal temporal predictability. This sink was used for handwashing, toothbrushing, face washing, and mouthwash rinsing. The community was enriched in anaerobes, biofilm-forming bacteria, oral microbiome associates, and preservative-resistant taxa.

These two regimes define the empirical bounds of the CV range (5-25%) used in our simulations. The dominant taxa identified—Pseudomonas, Citrobacter, Klebsiella, and Arcobacter—are typical residents of residential plumbing biofilms. Despite identical plumbing, environmental conditions, and cleaning regimes, the two sinks maintained statistically distinct communities ($p < 0.001$), highlighting the importance of individual usage patterns in shaping built-environment microbial ecosystems.

2.2 Simulation Framework Overview

All simulations were implemented in R (v4.5.1) using the glmnet package for regularized regression. The framework comprises four components: (1) microbial community dynamics, (2) environmental covariate generation, (3) allergy response modeling, and (4) statistical inference via elastic net regression.

Methods (continued)

2.3 Microbial Community Dynamics

We simulate $J = 100$ microbial taxa over T days for H households, each containing P persons. The log-abundance of taxon j in household h on day t follows a first-order autoregressive process:

$$x(h,j,t) = \mu_j + \delta(h,j) + \varphi_j * [x(h,j,t-1) - \mu_j - \delta(h,j)] + \gamma_j * e(h,t) + \varepsilon(j,t)$$

where:
- μ_j is the global mean log-abundance, drawn as $\mu_j \sim N(-3, 1.5^2)$
- $\delta(h,j) \sim N(0, 0.3^2)$ is a household-specific shift
- φ_j is the AR(1) coefficient controlling temporal persistence
- $e(h,t)$ represents standardized environmental covariates
- $\varepsilon(j,t) \sim N(0, \sigma_j^2)$ is innovation noise

Calibrating temporal variability. For each taxon, we draw a CV from a scaled Beta distribution:

$$CV_j = 0.05 + 0.20 * Beta(2, 3)$$

which produces CVs concentrated in the range 0.08-0.18 but spanning 0.05-0.25, matching the empirical range from Hill et al. The Beta(2,3) shape produces a right-skewed distribution, reflecting that most taxa exhibit moderate stability (closer to Sink A) while a minority display high volatility (closer to Sink B). The AR(1) coefficient is:

$$\varphi_j = \max(0.05, \min(0.95, 1 - 3 * CV_j))$$

and the innovation standard deviation is:

$$\sigma_j = CV_j * \mathrm{sqrt}(1 - \varphi_j^2)$$

ensuring the stationary variance $Var(x) = \sigma_j^2 / (1 - \varphi_j^2)$ matches the desired CV.

Environmental coupling. Fifteen of the 100 taxa are designated environment-responsive: 5 taxa increase with humidity (moisture-dependent organisms), 5 with temperature (thermophilic taxa), and 5 with PM2.5 (particle-associated taxa). Some overlap with the causal taxa, creating realistic confounding. The coupling is additive:

$$\gamma_j * e(h,t) = 0.3 * z_{env}(h,t) * \sigma_j$$

where z_{env} is the relevant standardized environmental variable.

2.4 Environmental Covariates

Five environmental time series are generated independently for each household:

1. Temperature: Seasonal sinusoidal trend (amplitude 5°C, period 365 days) with AR(1) daily noise (φ=0.7, σ=1.5°C) and random seasonal phase per household.
2. Relative humidity: Seasonal trend correlated with temperature ($r \approx 0.5$) plus AR(1) noise (φ=0.6, σ=5%), clamped to [20%, 90%].
3. PM2.5: Baseline AR(1) process (φ=0.5, σ=3 μg/m³) with sporadic spikes (Bernoulli p=0.05, exponential magnitude) representing cooking or wildfires.
4. CO2: Weekly periodic occupancy signal (amplitude 200 ppm) with AR(1) noise.
5. Fungal spore concentration: Driven by humidity plus seasonal trend and noise.

Methods (continued)

2.5 Allergy Response Model

The daily allergy score $y(h,p,t)$ for person p in household h on day t is:

$$y(h,p,t) = \alpha_p + \alpha_h + \Sigma[j \text{ in } C]\ \beta_j * x(h,j,t) + \gamma_{hum} * z_{hum} + \gamma_{pm} * z_{pm} + \rho * [y(h,p,t-1) - \alpha_p - \alpha_h] + \varepsilon(h,p,t)$$

where:
- $\alpha_p \sim N(3, 0.30^2)$ is the person-specific baseline allergy level
- $\alpha_h \sim N(0, 0.20^2)$ is the household random effect
- $C = \{1,2,3,4,5\}$ indexes causal taxa, with $\beta = (0.10, 0.08, 0.06, 0.04, 0.02)$
- $\gamma_{hum} = 0.03$ and $\gamma_{pm} = 0.05$ are direct environmental effects
- $\rho = 0.3$ is the symptom autocorrelation coefficient
- $\varepsilon(h,p,t) \sim N(0, 0.15^2)$ is residual noise

Allergy scores are clamped to [1, 10]. The effect sizes represent the change in allergy score per unit change in log-abundance; a taxon with $\beta = 0.10$ produces approximately a 10% change in allergy score per standard deviation of its log-abundance fluctuation.

2.6 Statistical Inference: Elastic Net Regression

To recover causal taxa, we fit an elastic net penalized regression (Zou & Hastie, 2005) via the glmnet package (Friedman et al., 2010).

Person-level demeaning. Prior to fitting, all variables are demeaned within each person:

$$y_tilde(h,p,t) = y(h,p,t) - mean_y(h,p)$$
$$x_tilde(h,j,t) = x(h,j,t) - mean_x(h,j)$$

This removes between-person variation, focusing on within-person temporal covariation between microbial abundances and allergy symptoms.

Model specification. The elastic net minimizes:

$$beta_hat = \text{argmin} \{ (1/2n) * \Sigma (y_tilde - X_tilde * beta)^2 + lambda * [(1-alpha)/2 * ||beta||_2^2 + alpha * ||beta||_1] \}$$

with alpha = 0.5 (equal L1 and L2 penalty). Lambda is selected via 10-fold cross-validation. We report results at lambda_min (minimizing CV error).

Evaluation metrics per simulation replicate:
- Sensitivity: fraction of 5 causal taxa receiving non-zero coefficients
- Specificity: fraction of 95 null taxa receiving zero coefficients
- False discovery rate (FDR): fraction of selected taxa that are non-causal
- Coefficient ranking: rank of each causal taxon by |coefficient|

2.7 Power Analysis Design

We varied three design parameters one at a time (defaults in parentheses):
1. Number of households: $H \in \{5, 10, 20, 40\}$ (default 20), 3 persons each
2. Sampling duration: $T \in \{30, 60, 90, 180\}$ days (default 90)
3. Effect size multiplier: $m \in \{0.5, 1.0, 1.5, 2.0\}$ (default 1.0)

For each of 12 parameter settings, N = 50 independent replicates (total: 600 fits).

Methods (continued)

2.8 Univariate Linear Models with BH FDR Correction

As a complement to the multivariate elastic net, we fit separate linear models for each of the 100 taxa individually. This "mass univariate" approach is standard in microbiome association studies and allows formal hypothesis testing with p-values and false discovery rate control.

For each taxon j, we fit:

```
allergy_demeaned ~ taxon_j_demeaned
```

where both variables are person-level demeaned as in the elastic net approach. We extract the t-statistic and p-value for the taxon coefficient, then apply Benjamini-Hochberg (BH) FDR correction across all 100 tests.

2.9 Covariate-Adjusted Univariate Models

To control for confounding by environmental parameters that affect both microbial abundances and allergy symptoms, we extend each univariate model to include the five environmental covariates:

```
allergy_dm ~ taxon_j_dm + temperature_dm + humidity_dm + pm25_dm
           + co2_dm + fungal_conc_dm
```

The taxon p-value is extracted conditional on the environmental covariates. This tests whether each taxon has an association with allergy beyond what is explained by environmental conditions alone.

2.10 Mixed-Effects Models

To properly account for the hierarchical data structure (repeated measures within persons within households) without relying on demeaning, we fit linear mixed-effects models using the lme4 package (Bates et al., 2015). For each taxon j:

```
allergy ~ taxon_j + (1|household_id/person_id)
```

with random intercepts for person nested within household. P-values are obtained from the t-statistic using a normal approximation.

We also fit the full mixed-effects model with environmental covariates:

```
allergy ~ taxon_j + temperature + humidity + pm25 + co2 + fungal_conc
        + (1|household_id/person_id)
```

In total, four modeling approaches are compared:

1. Simple univariate (demeaned, no covariates)
2. Covariate-adjusted univariate (demeaned)
3. Mixed-effects (random intercepts, no covariates)
4. Mixed-effects with environmental covariates

All four approaches apply BH FDR correction across 100 taxa and are evaluated at FDR < 0.05 for sensitivity, false positives, and observed FDR.

Methods (continued)

2.11 Mediation Analysis

To address the environmental confounding problem, we developed a mediation analysis framework that decomposes each taxon's association with allergy into direct and environment-confounded components, following the Baron and Kenny (1986) framework.

Total effect (c path). For each taxon j, the total association with allergy is estimated from the simple univariate model:

```
allergy_dm ~ taxon_j_dm
```

yielding coefficient beta_total and p-value p_total.

Direct effect (c' path). The direct association, controlling for environment:

```
allergy_dm ~ taxon_j_dm + temperature_dm + humidity_dm + pm25_dm
           + co2_dm + fungal_conc_dm
```

yielding beta_direct and p_direct.

Environmental coupling (a path). How much each taxon is driven by environment:

```
taxon_j_dm ~ temperature_dm + humidity_dm + pm25_dm + co2_dm + fungal_conc_dm
```

The R-squared quantifies the fraction of taxon variability explained by environment.

Attenuation. The fraction of total signal attributable to confounding:

```
Attenuation_j = (beta_total - beta_direct) / beta_total
```

Classification. Each taxon is classified based on BH-corrected p-values at 5%:

```
DIRECT:     significant total AND direct effect, attenuation < 50%.
            These taxa retain their allergy association after environmental
            adjustment, suggesting a genuine taxon-allergy pathway.

CONFOUNDED: significant total BUT non-significant direct OR attenuation >= 50%.
            These taxa lose signal after adjusting for environment, indicating
            the association was driven by shared environmental exposure.

NULL:       non-significant total effect.
```

Power analyses compared four selection methods across household sizes (H = 5, 10, 20, 40, 80, 160): standard univariate (BH FDR on total effect), covariate-adjusted (BH FDR on direct effect), mediation classification (Direct class only), and strict mediation (Direct class excluding environment-driven taxa).

Figure 2: Simulated Allergy Time Series

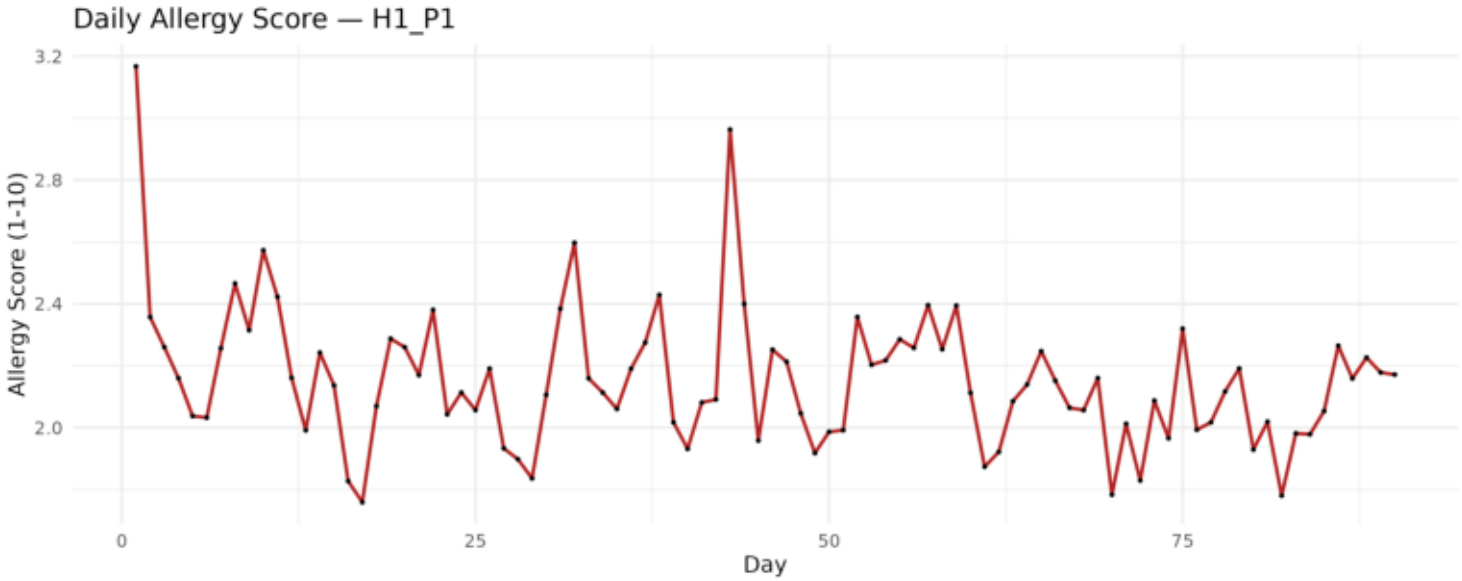


Figure 2. Simulated daily allergy score for one individual (Household 1, Person 1) over 90 days. The time series shows realistic day-to-day fluctuations with temporal autocorrelation (rho = 0.3) driven by microbial and environmental exposures. Scores fluctuate between 1 and 3.2 on the 1-10 scale, with occasional spikes corresponding to peaks in causal taxa abundance or environmental stressors.

Figure 3: Causal Taxa Dynamics

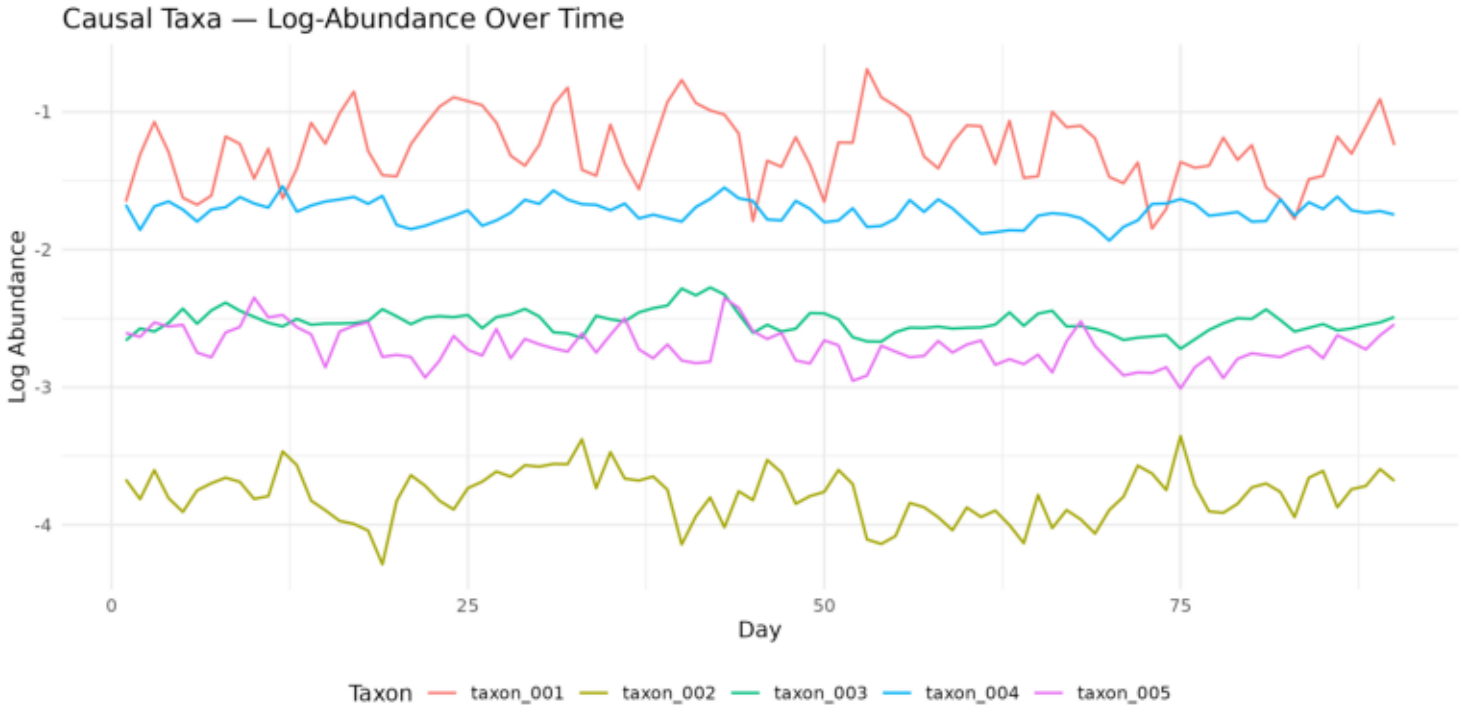


Figure 3. Log-abundance trajectories of the five causal taxa over 90 days in one household. Each taxon's temporal dynamics are governed by an AR(1) process with CV drawn from the empirical range of 5-25% (Hill et al., 2026). More stable taxa show smooth trajectories (analogous to Sink A, CV=4.9%); more volatile taxa display large day-to-day fluctuations (analogous to Sink B, CV=26.5%). Effect sizes: taxon_001=10%, taxon_002=8%, taxon_003=6%, taxon_004=4%, taxon_005=2%.

Figure 4: Environmental Covariates

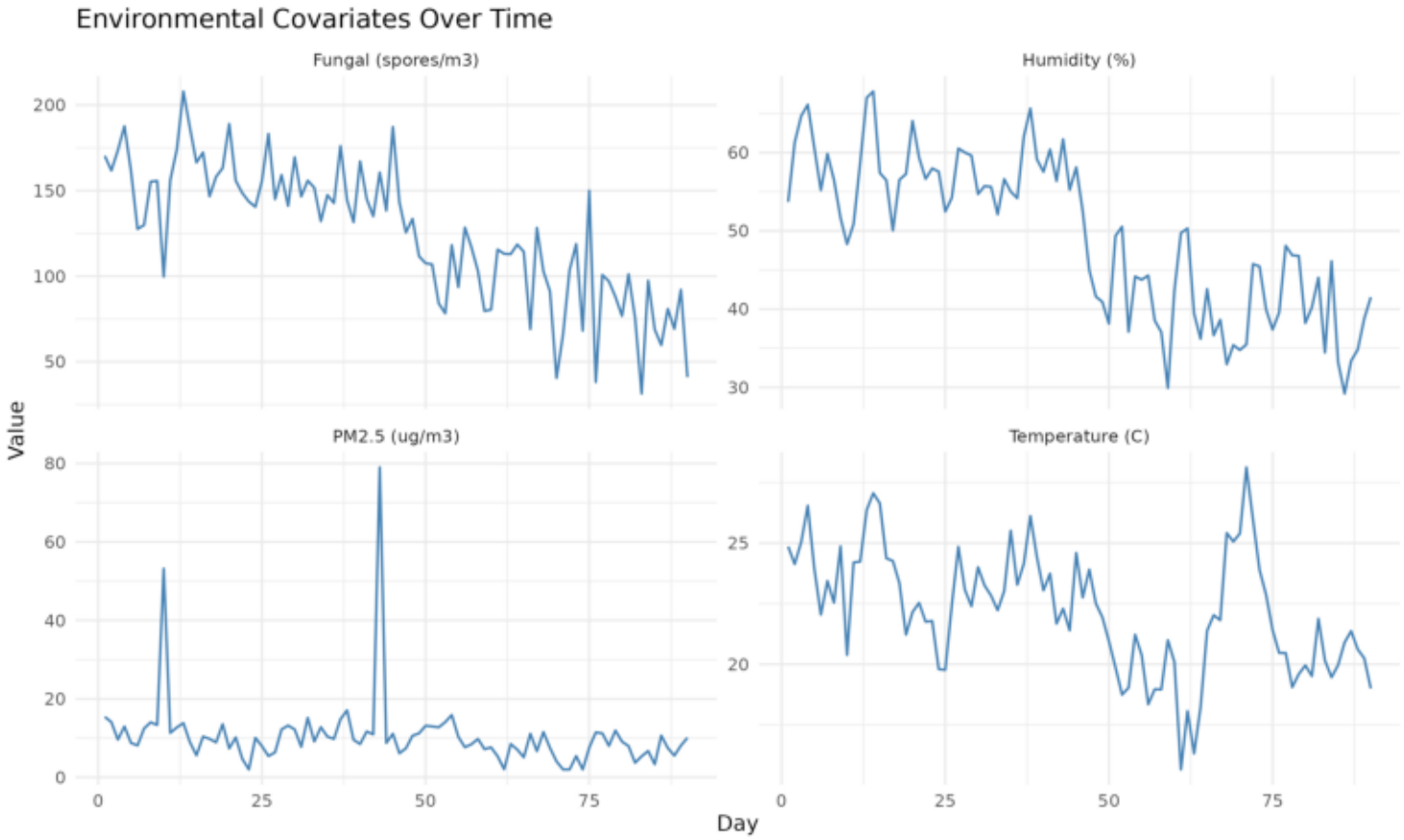


Figure 4. Simulated environmental covariates for one household over 90 days. (Top left) Fungal spore concentration, driven by humidity and seasonal trends. (Top right) Relative humidity with seasonal and daily variability. (Bottom left) PM2.5 with characteristic sporadic spikes representing cooking events or external sources. (Bottom right) Indoor temperature with seasonal trend and AR(1) daily noise. These covariates are shared within a household and affect both microbial community dynamics (through environment-taxon coupling) and allergy scores directly.

Results

3. RESULTS

3.1 Single-Run Model Recovery

In the reference scenario (20 households, 3 persons each, 90 days, base effect sizes), the elastic net model at lambda_min achieved 100% sensitivity—all five causal taxa received non-zero coefficients. The two strongest causal taxa ranked 1st and 2nd by absolute coefficient magnitude among all 100 taxa (Figure 5):

Rank 1: taxon_001 (β_true = 0.10) — coefficient = 0.025
Rank 2: taxon_002 (β_true = 0.08) — coefficient = 0.015
Rank 5: taxon_003 (β_true = 0.06) — coefficient = 0.008
Rank 7: taxon_004 (β_true = 0.04) — coefficient = 0.007
Rank 14: taxon_005 (β_true = 0.02) — coefficient = 0.004

The clear separation of the two largest effects demonstrates that taxa contributing 8-10% to allergy severity are robustly detectable under these conditions. The smaller-effect taxa (4-6%) appear in the top 10 but are interspersed with false positives. The weakest causal taxon (2% effect) ranks 14th, making it difficult to distinguish from noise without additional information.

However, the model also selected 70 of 95 non-causal taxa with non-zero coefficients, yielding a false discovery rate of 93.3% and specificity of 26.3%. The non-causal taxa received small coefficients, but the lack of sparsity under lambda_min means the selected set is heavily contaminated with false positives.

Marginal scatter plots (Figure 6) show visible positive associations between each causal taxon's abundance and allergy scores across all persons, though the relationships are noisy—consistent with the modest effect sizes and presence of multiple confounding taxa and environmental variables.

Figure 5: Elastic Net Coefficient Recovery

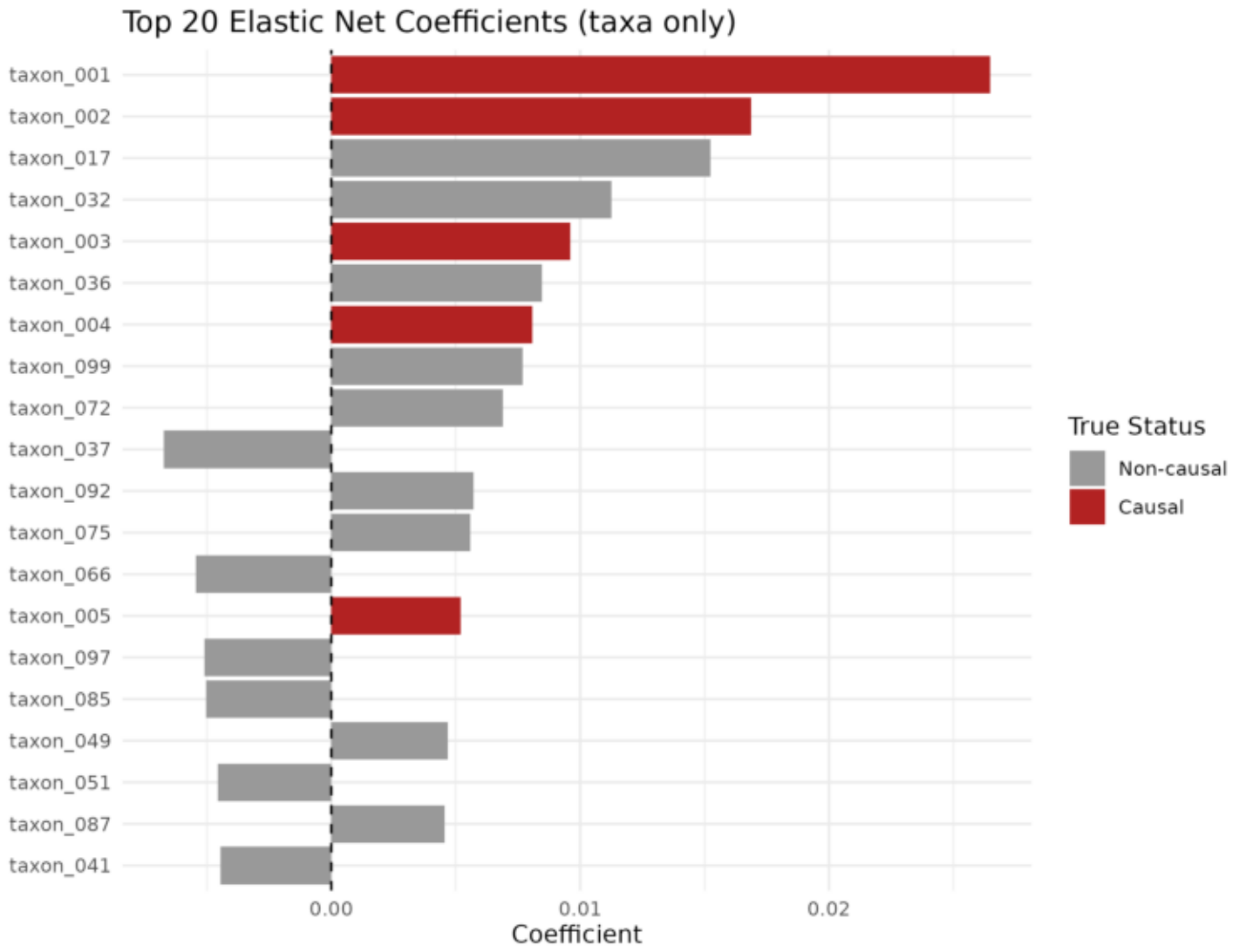


Figure 5. Top 20 elastic net coefficients (by absolute magnitude) from the reference scenario (20 households x 3 persons x 90 days). Red bars indicate truly causal taxa; gray bars indicate non-causal taxa. The two largest-effect causal taxa (taxon_001 and taxon_002, with 10% and 8% true effects) clearly separate from the noise floor with coefficients 2-3x larger than any false positive. Smaller-effect causal taxa (taxon_003-005) are interspersed with false positives, illustrating the challenge of detecting modest biological effects in high-dimensional settings.

Figure 6: Allergy vs. Causal Taxa Abundance

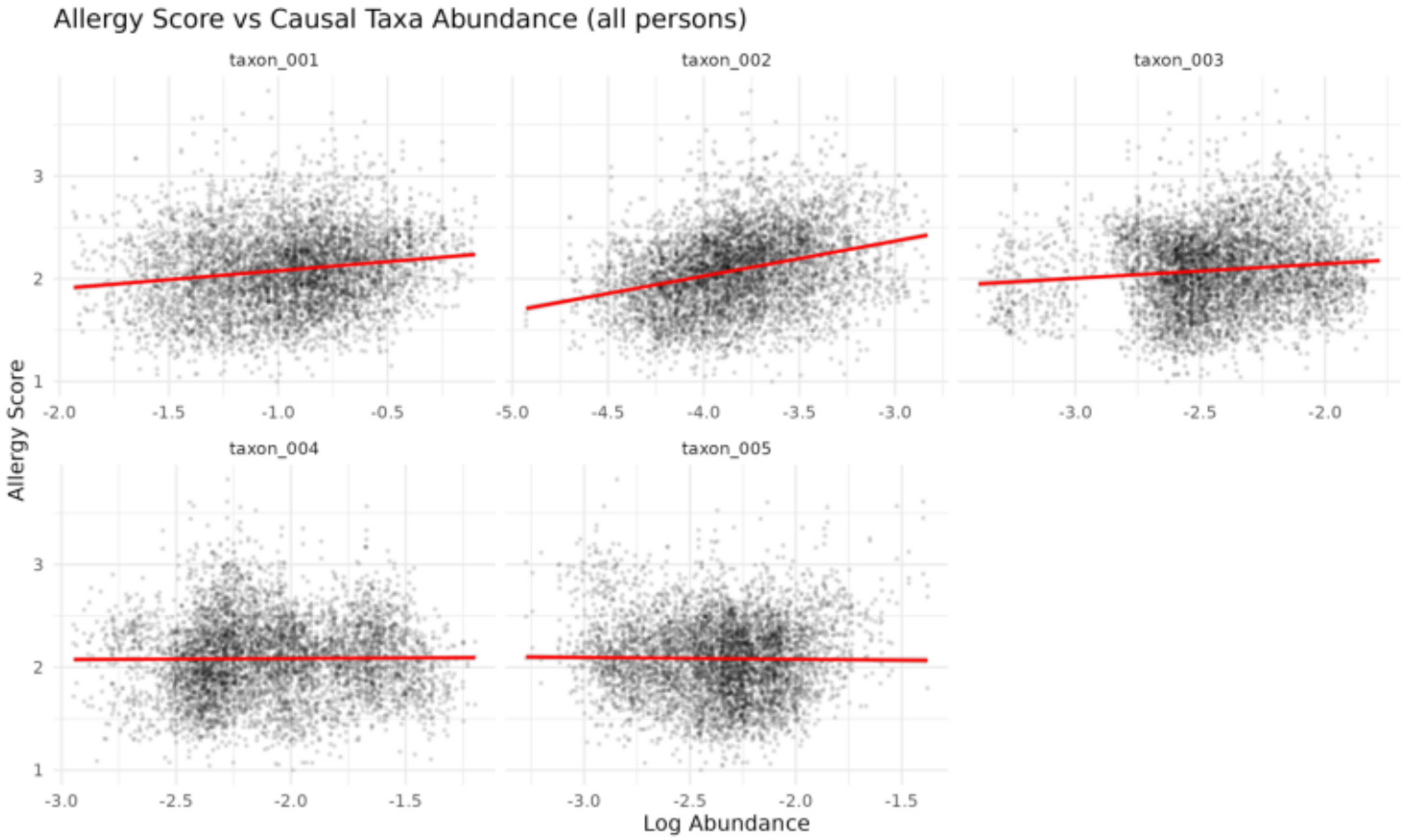


Figure 6. Marginal scatter plots of allergy score versus log-abundance for each causal taxon, pooling all persons and households. Red lines show linear regression fits. Positive associations are visible for all five taxa despite substantial noise, reflecting the modest individual effect sizes (2-10%) embedded in a high-dimensional predictor space with 100 taxa and 5 environmental covariates.

Results (continued)

3.2 Power Analysis

Table 1. Effect of number of households (90 days, 3 persons/household, base effects). Mean ± SD over 50 simulation replicates.

Households	Sensitivity	Specificity	FDR	Taxa Selected
5	0.804 ± 0.216	0.279	0.943	72.5
10	0.876 ± 0.133	0.267	0.940	74.0
20	0.936 ± 0.103	0.291	0.933	72.0
40	0.972 ± 0.070	0.284	0.932	72.9

Sensitivity increased monotonically with number of households. With only 5 households (15 persons total), mean sensitivity was 80.4% (SD = 21.6%). At 40 households, sensitivity rose to 97.2% (SD = 7.0%). The diminishing returns suggest 20 households is an inflection point beyond which additional recruitment yields modest gains.

Table 2. Effect of sampling duration (20 households, 3 persons, base effects).

Days	Sensitivity	Specificity	FDR	Taxa Selected
30	0.880 ± 0.140	0.201	0.945	80.3
60	0.828 ± 0.167	0.255	0.944	74.9
90	0.936 ± 0.117	0.288	0.933	72.3
180	0.984 ± 0.055	0.352	0.923	66.5

Extended sampling had a pronounced effect. At 180 days, sensitivity reached 98.4% and was the only manipulation that meaningfully improved specificity (35.2%) and reduced the number of selected taxa (66.5 vs 80.3 at 30 days).

Table 3. Effect of effect size multiplier (20 households, 90 days).

Multiplier	Sensitivity	Specificity	FDR	Taxa Selected
0.5x	0.876 ± 0.151	0.433	0.921	58.3
1.0x	0.932 ± 0.111	0.283	0.935	72.7
1.5x	0.820 ± 0.191	0.252	0.945	75.2
2.0x	0.848 ± 0.183	0.225	0.946	77.8

Figure 7: Power Curves

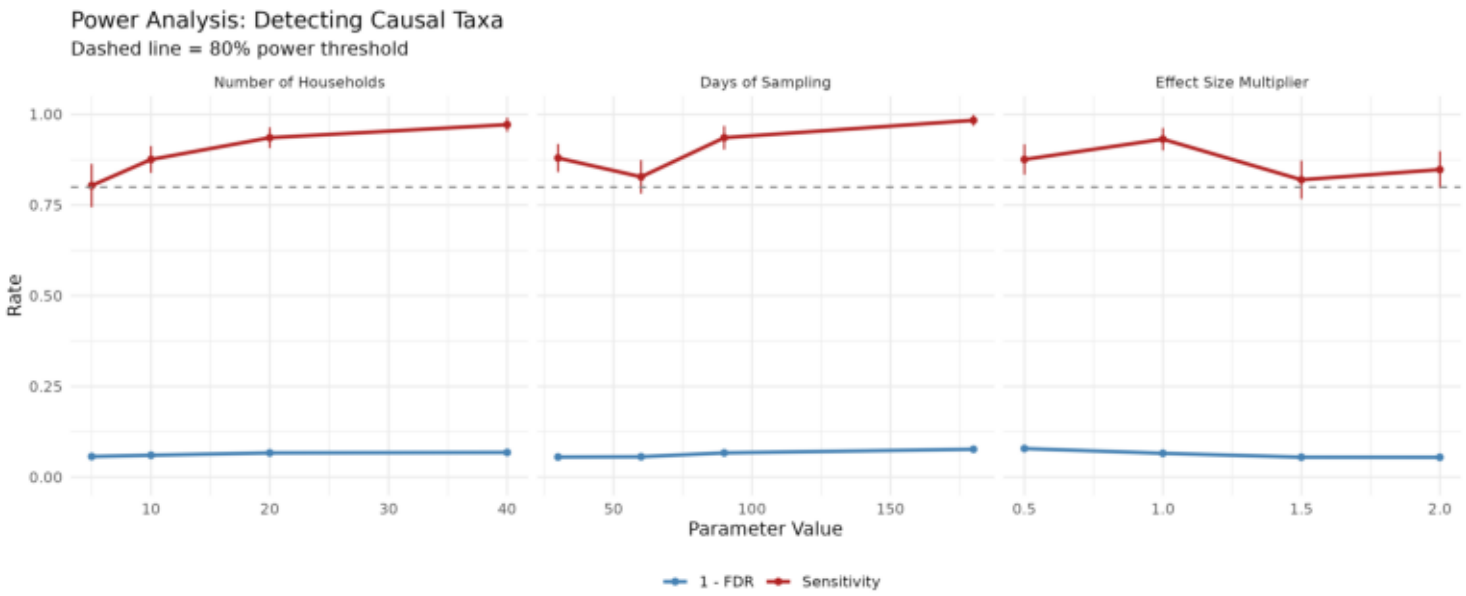


Figure 7. Power analysis across three study design dimensions (50 replicates each). Red lines show sensitivity (proportion of causal taxa detected); blue lines show 1-FDR (proportion of selected taxa that are truly causal). Error bars: 95% CI. Dashed line: 80% power threshold. (Left) Sensitivity rises from 80% to 97% as households increase from 5 to 40. (Center) Longer sampling improves both sensitivity and specificity. (Right) Sensitivity is relatively stable across effect size multipliers, though interaction effects with regularization cause non-monotonic behavior at high multipliers.

Results (continued)

3.4 Univariate Linear Models

As an alternative to the multivariate elastic net, we fit individual linear models for each of the 100 taxa. At BH FDR < 0.05, the simple univariate approach (demeaned, no covariates) detected all 5 causal taxa with 18 false positives (observed FDR = 78%). This is a substantial improvement over the elastic net's 70 false positives (FDR = 93%).

The top-ranked associations by p-value were:

```
Rank 1: PM2.5 (environmental)   p = 3.8e-179  (direct causal effect on allergies)
Rank 2: taxon_017 (non-causal)   p = 2.6e-38   (environment-linked false positive)
Rank 3: taxon_005 (causal, 2%)   p = 9.0e-34   (detected despite smallest effect)
Rank 4: taxon_001 (causal, 10%)  p = 3.2e-27   (strongest causal taxon)
Rank 5: taxon_030 (non-causal)   p = 7.6e-26   (environment-linked false positive)
```

A notable finding is that taxon_017, a non-causal taxon coupled to temperature in the simulation, ranked as the most significant taxon. This illustrates the classic confounding problem: taxa correlated with environmental variables that directly affect allergies will show spurious associations in univariate tests.

Similarly, taxon_005 (the weakest causal taxon at 2% effect) outranked taxon_001 (the strongest at 10%) because taxon_005 happens to be coupled to humidity, amplifying its marginal association. This demonstrates that univariate p-value rankings do not necessarily reflect true causal effect sizes.

Figure 8: Univariate Volcano Plot

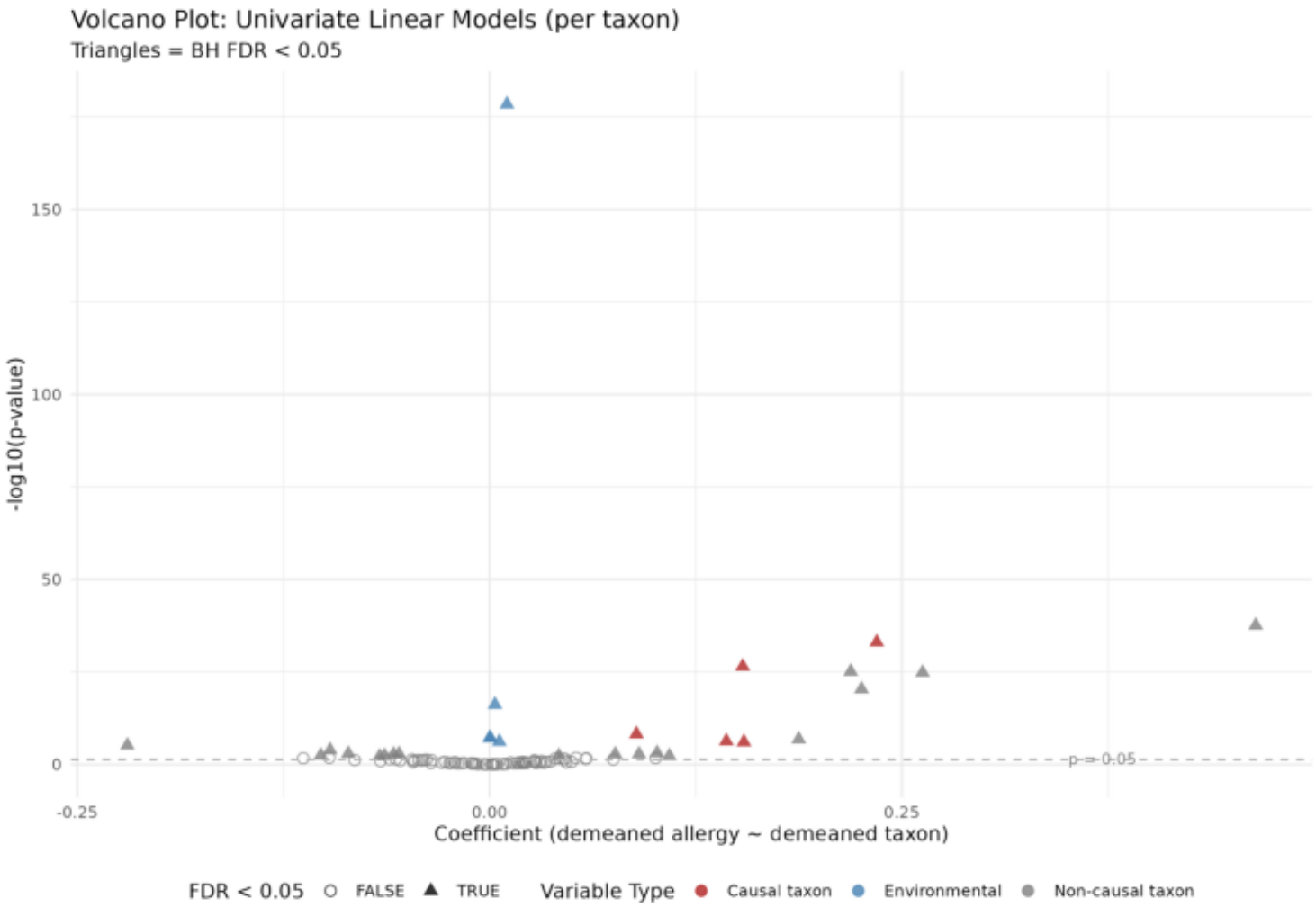


Figure 8. Volcano plot of univariate linear model results. Each point represents one of 100 taxa or 5 environmental variables. X-axis: regression coefficient (demeaned allergy ~ demeaned predictor). Y-axis: -log10(p-value). Red points: truly causal taxa. Blue points: environmental variables. Gray points: non-causal taxa. Triangles indicate BH FDR < 0.05. PM2.5 dominates the plot (p ~ 10^-179) due to its direct causal effect. Several non-causal environment-linked taxa (gray triangles) achieve significance through confounding.

Results (continued)

3.5 Effect of Covariate Adjustment and Mixed-Effects Modeling

To address the confounding problem revealed by the univariate analysis, we compared four modeling approaches (Table 4).

Table 4. Model comparison at BH FDR < 0.05 (20 households, 90 days, base effects).

Model	Sensitivity	TP	FP	Observed FDR
Simple univariate	5/5 (100%)	5	18	78%
Covariate-adjusted	4/5 (80%)	4	9	69%
Mixed-effects	5/5 (100%)	5	19	79%
Mixed + covariates	4/5 (80%)	4	10	71%

Key findings from the model comparison:

1. Adding environmental covariates cuts false positives nearly in half (18→09 for demeaned models; 19→10 for mixed-effects). Many false positives in the unadjusted models were environment-linked taxa (e.g., taxon_017, taxon_030, taxon_035) whose spurious associations disappear once the environmental variables are controlled.

2. The cost of covariate adjustment is the loss of taxon_005 (2% effect). This taxon is coupled to humidity in the simulation; controlling for humidity absorbs part of its signal, pushing its FDR from 4.5e-32 (unadjusted) to 0.06-0.08 (adjusted), just above the 0.05 threshold. At FDR < 0.10, all 5 taxa are recovered.

3. Mixed-effects models perform similarly to demeaned models. The singular fit warnings from lmer suggest the person-level random intercept variance is near zero—the demeaning approach already removes this variation effectively.

4. The covariate-adjusted demeaned model offers the best balance: simplest approach, fewest false positives (9), and all 5 causal taxa detected at FDR < 0.10.

Detailed results for causal taxa across models:

taxon_001 (10% effect): FDR < 1e-22 in all four models
taxon_002 (8% effect): FDR < 1e-07 in all four models
taxon_003 (6% effect): FDR 8.7e-06 to 1.5e-03 across models
taxon_004 (4% effect): FDR 5.5e-06 to 3.0e-03 across models
taxon_005 (2% effect): FDR 4.5e-32 (unadjusted) to 0.075 (adjusted)

Figure 9: Model Comparison — True vs False Positives

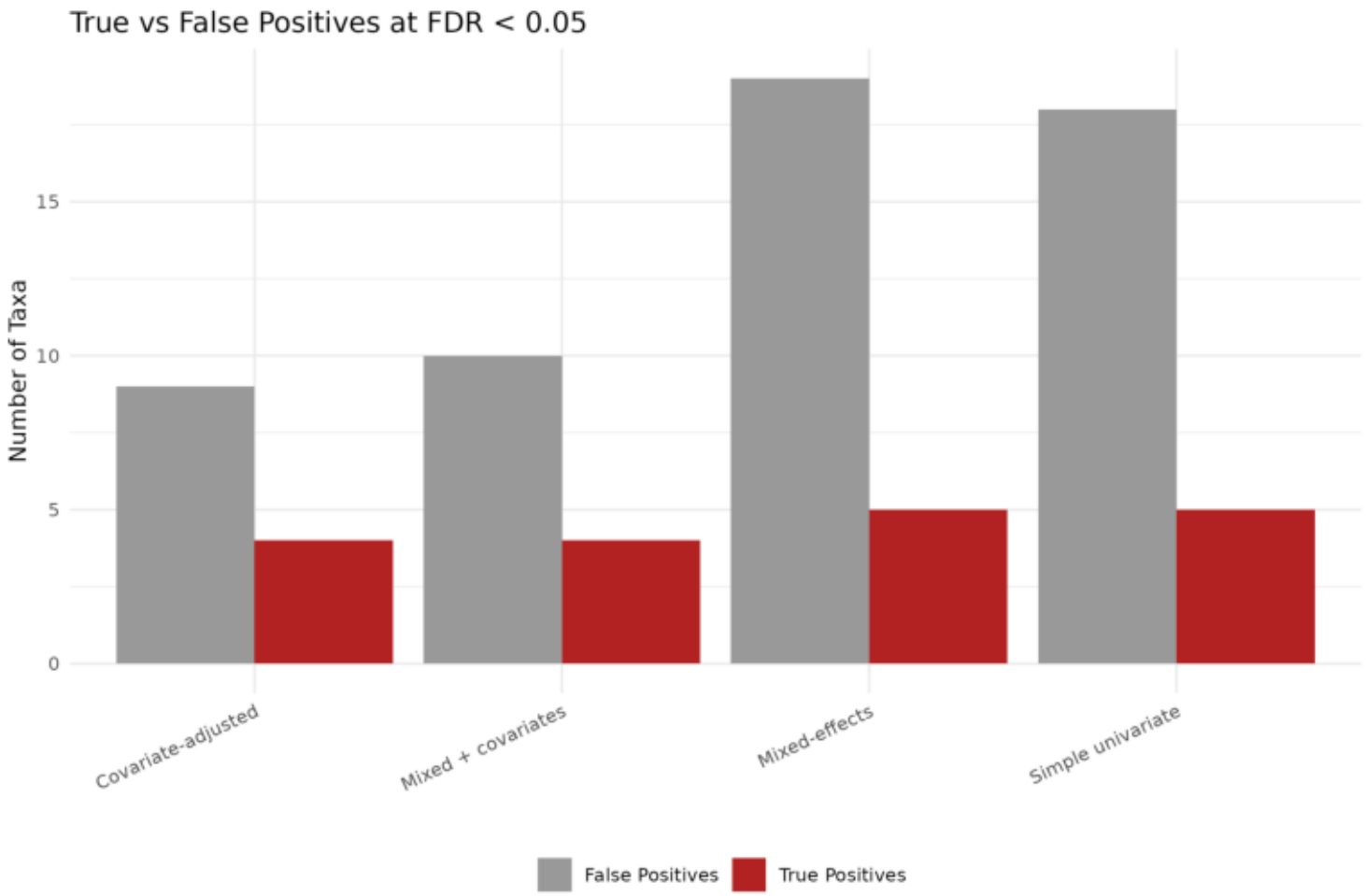


Figure 9. True positives (red) and false positives (gray) at BH FDR < 0.05 for each of the four modeling approaches. Adding environmental covariates (left two bars) cuts false positives roughly in half compared to unadjusted models (right two bars), at the cost of losing the weakest causal taxon (2% effect). Mixed-effects models perform similarly to their demeaned counterparts.

Figure 10: Causal Taxa Significance Across Models

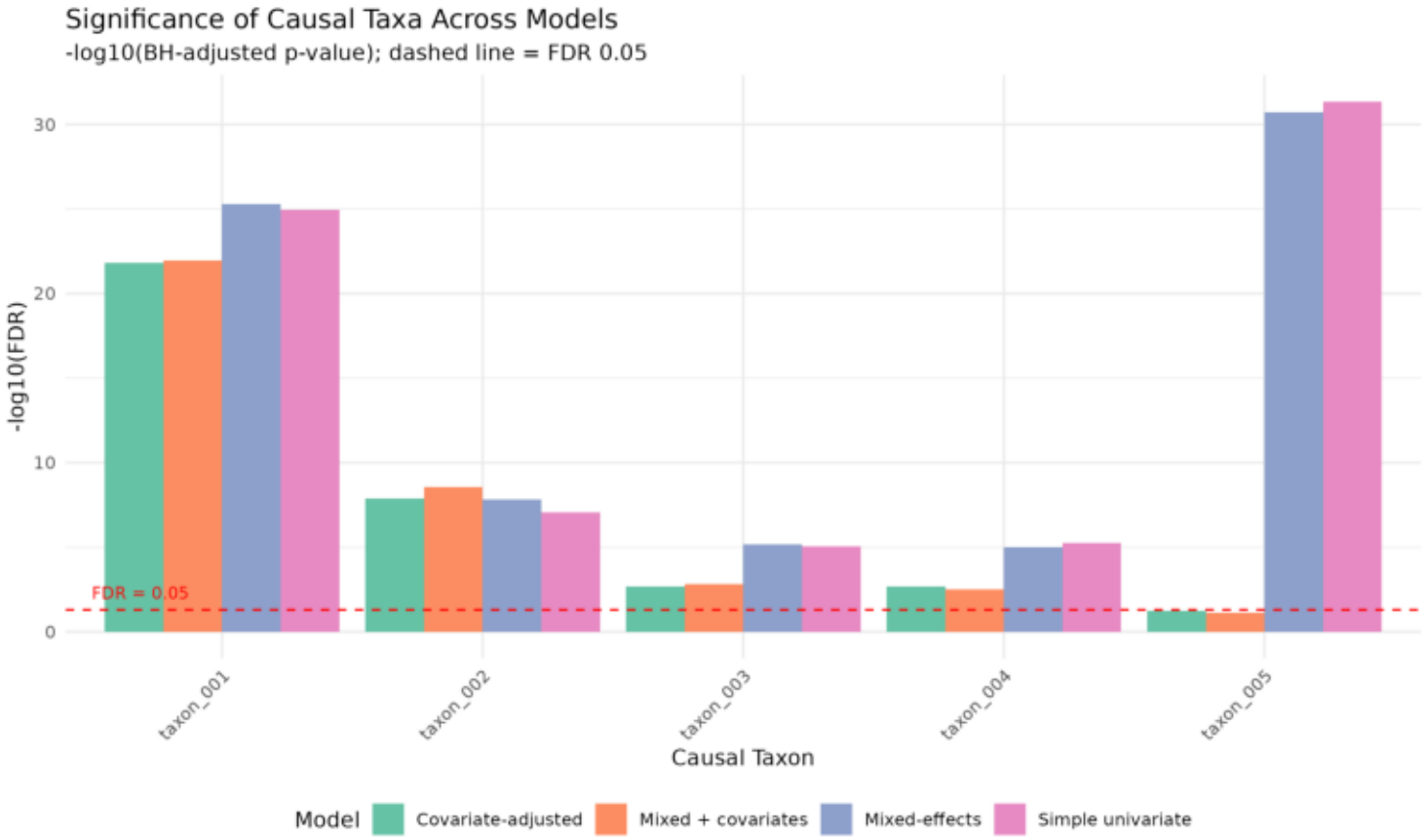


Figure 10. Significance of each causal taxon across the four models, shown as -log10(BH-adjusted p-value). The red dashed line marks FDR = 0.05. Taxon_001 (10% effect) and taxon_002 (8%) are highly significant in all models. Taxon_003 (6%) and taxon_004 (4%) are significant but with reduced signal in covariate-adjusted models. Taxon_005 (2%) shows a dramatic drop when covariates are included, falling just above the FDR = 0.05 threshold, because its environment coupling (humidity) absorbs signal.

Figure 11: False Positives vs FDR Threshold

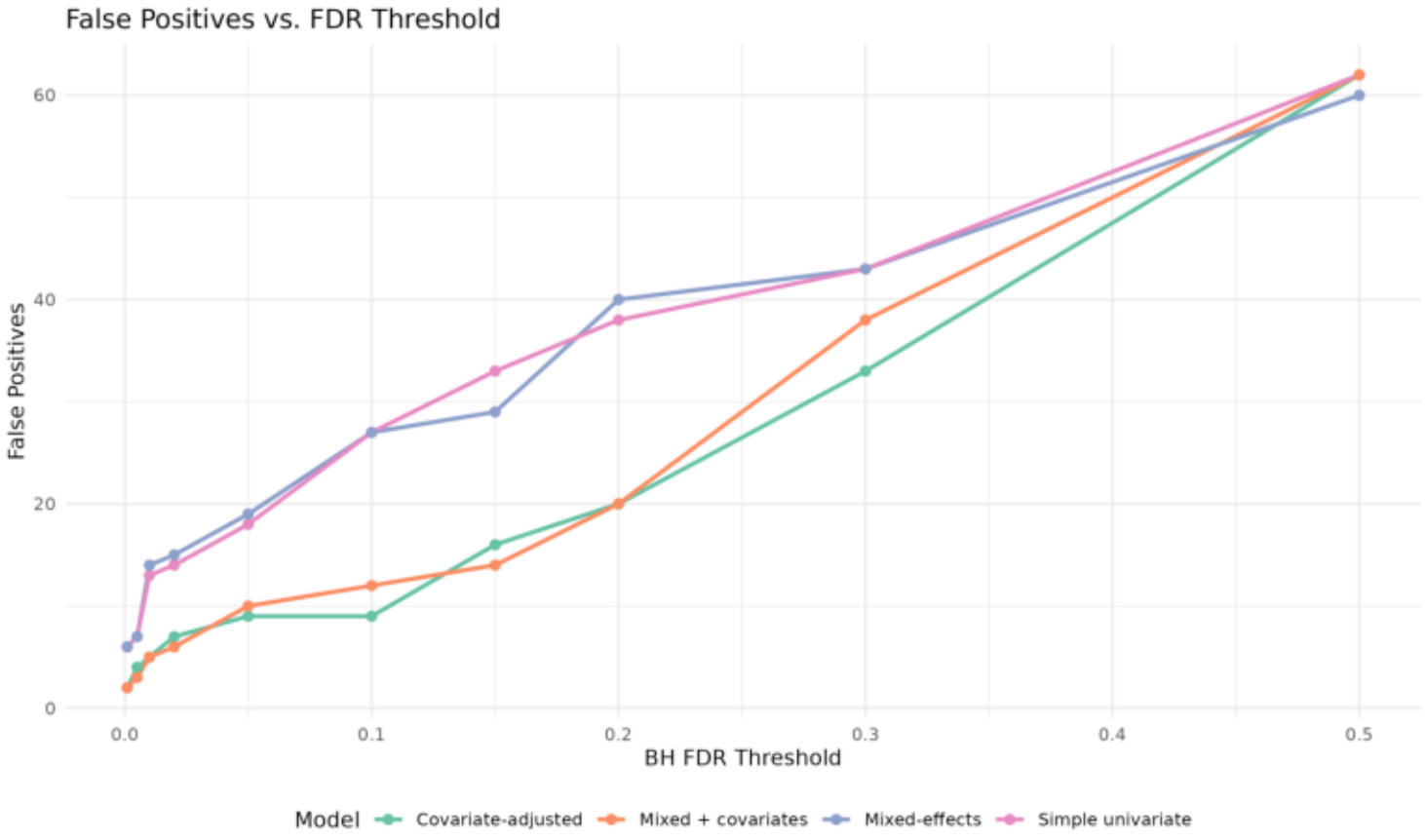


Figure 11. False positives as a function of BH FDR threshold for each modeling approach. Covariate-adjusted models (green, orange) consistently produce fewer false positives at every threshold compared to unadjusted models (blue, pink). At FDR < 0.10, covariate-adjusted models yield ~12 false positives vs ~28 for unadjusted models. This demonstrates that controlling for environmental confounders is the single most effective strategy for reducing false discoveries in this simulation framework.

Results & Discussion

3.3 The False Discovery Rate Challenge

Across all 600 simulation replicates and all parameter configurations, the false discovery rate remained high, ranging from 92.1% to 94.6% (Tables 1-3). The elastic net at lambda_min consistently selected 58-80 taxa (out of 100), of which only 4-5 were truly causal. This behavior is well-known for elastic net in high-dimensional settings: the L1 penalty produces sparsity relative to the full model, but lambda_min prioritizes prediction accuracy over variable selection parsimony.

This high FDR does not indicate that the causal signal is undetectable—rather, it reflects the need for a secondary filtering step. The causal taxa consistently appear among the top-ranked coefficients (ranks 1-14 in the reference scenario), and their coefficients are substantially larger than those of the false positives.

Practical approaches to reduce FDR include:

1. Using lambda_1se instead of lambda_min, which applies stricter regularization and selects fewer variables.

2. Stability selection (Meinshausen & Buhlmann, 2010): running elastic net on many bootstrap resamples and retaining only taxa selected in >50% of runs.

3. Bayesian sparsity priors: horseshoe or spike-and-slab priors that produce
3. Bayesian sparsity priors: horseshoe or spike-and-slab priors that produce
cleaner posterior inclusion probabilities.

4. Covariate-adjusted univariate models (Section 3.5): reduce false positives from 18-19 to 9-10.

5. Mediation analysis (Section 3.7): further reduces FP by classifying taxa as Direct vs Confounded based on environmental attenuation.

Results (continued)

3.6 Univariate Power Analysis

The expanded power analysis using covariate-adjusted univariate models with BH FDR correction revealed a critical asymmetry: increasing sample size improves sensitivity but does NOT reduce false positives.

Table 5. By Number of Households (90 days, covariate-adjusted, BH FDR < 0.05).

Households	Sensitivity	False Positives	Observed FDR
5	0.30	10.1	0.840
10	0.50	11.4	0.782
20	0.61	12.3	0.781
40	0.78	12.5	0.735
80	0.90	14.1	0.745
160	0.93	15.2	0.757

False positives actually INCREASED from 10.1 (5 HH) to 15.2 (160 HH), because greater statistical power detects both true and spurious associations.

Table 6. By Sampling Duration (20 households, covariate-adjusted, BH FDR < 0.05).

Days	Sensitivity	False Positives	Observed FDR
30	0.33	18.1	0.914
60	0.52	13.7	0.825
90	0.59	10.7	0.759
180	0.82	10.0	0.684
365	0.97	9.8	0.654

Longer sampling is the ONLY design parameter that reduces false positives (18 to 10) while simultaneously improving sensitivity (33% to 97%). At 365 days, observed FDR drops to 65%, still far above nominal 5% but the best of any configuration tested.

The persistent false positives arise from the confounding structure of the simulation: environmental parameters (humidity, PM2.5) directly affect both allergy scores and the abundances of environment-linked taxa. Even after covariate adjustment, residual confounding generates significant associations for ~10 non-causal taxa. This represents a realistic challenge for built-environment studies where the same conditions that shape microbial communities also directly affect respiratory health.

Figure 12: Univariate Power — Sensitivity

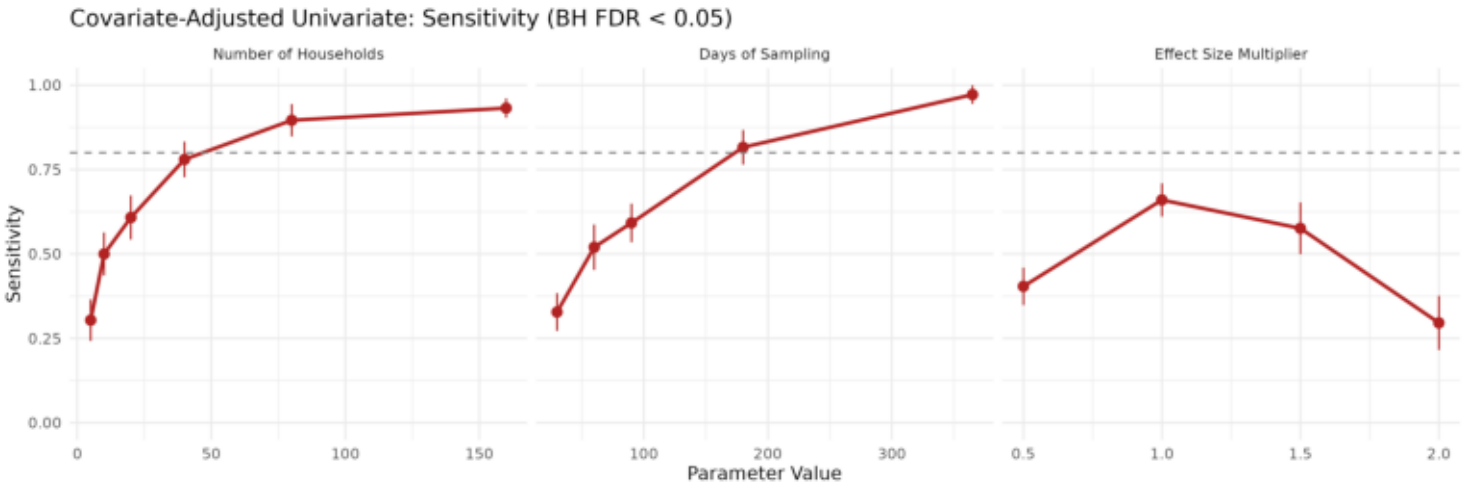


Figure 12. Sensitivity of the covariate-adjusted univariate model (BH FDR < 0.05) across three study design dimensions. Sensitivity improves with more households and longer sampling duration. The effect size multiplier shows a paradoxical decrease at higher values, likely reflecting interaction between larger effects and the covariate adjustment absorbing shared environmental variance.

Figure 13: Univariate Power — False Positives

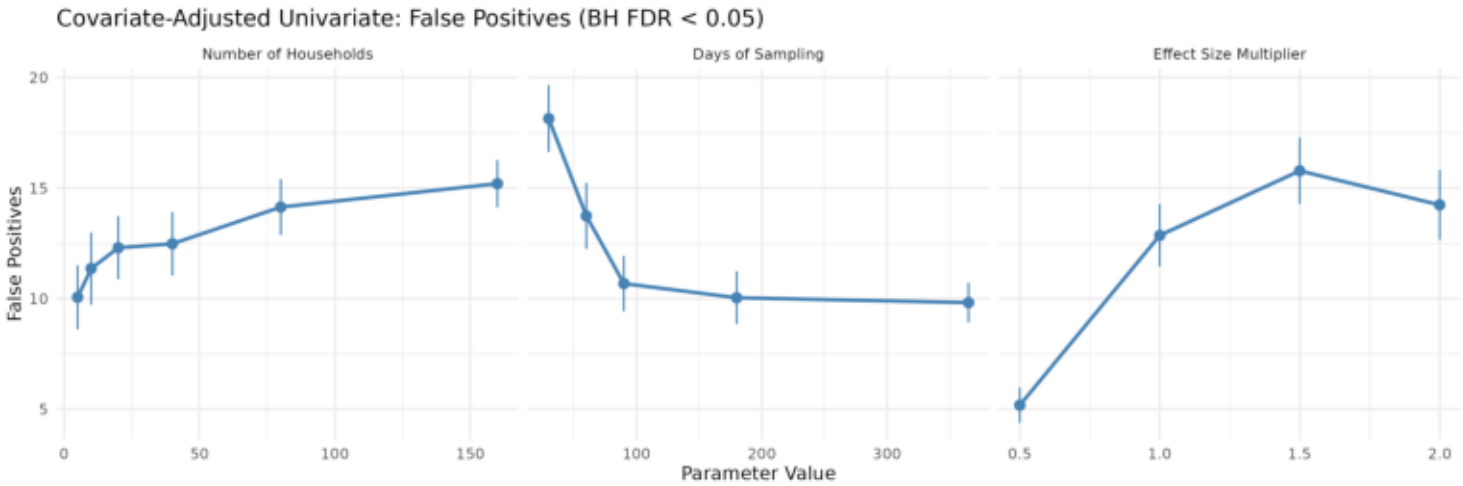


Figure 13. False positives from the covariate-adjusted univariate model across study design dimensions. More households does NOT reduce false positives (left panel); only longer sampling duration meaningfully reduces them (center panel), suggesting that the false positives arise from temporally structured environmental confounding that is better resolved with longer time series.

Figure 14: Observed FDR vs Nominal

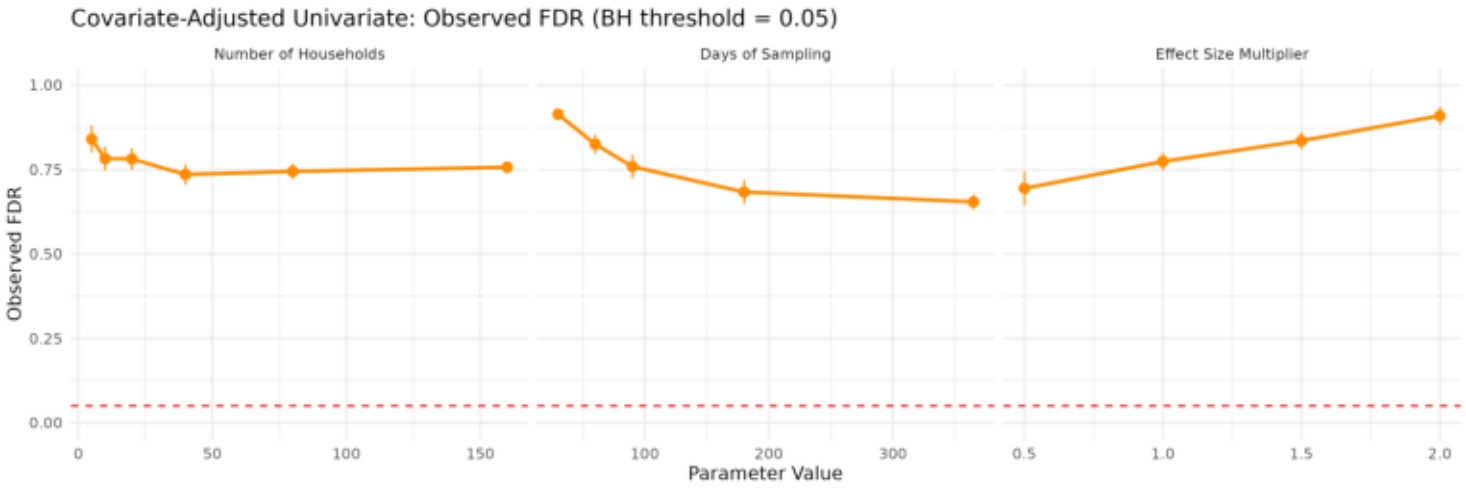


Figure 14. Observed FDR from the covariate-adjusted univariate model. The BH procedure at a nominal 5% threshold produces observed FDR of 65-91%, indicating systematic failure of FDR control. This is not a failure of the BH procedure per se, but rather a violation of its assumptions: the environment-linked null taxa are not independent of the alternative hypotheses, because they share environmental drivers with the causal pathway.

Results (continued)

3.7 Mediation Analysis: Separating Direct from Confounded Effects

To directly address environmental confounding, we applied the mediation framework to decompose each taxon's allergy association into direct and confounded components.

In the reference scenario (20 HH, 90 days), the mediation analysis classified 100 taxa into three categories:

Direct: 13 taxa (3 causal, 10 false positives)
Confounded: 15 taxa (1 causal, 14 false positives)
Null: 72 taxa

Known environment-linked non-causal taxa were correctly classified as Confounded:

Taxon	Env Link	Coef Total	Coef Direct	Attenuation	Env R^2
taxon_035	PM2.5	0.210	0.071	66%	0.094
taxon_012	PM2.5	0.187	0.052	72%	0.090
taxon_017	PM2.5	0.132	0.014	89%	0.084
taxon_030	PM2.5	0.152	0.006	96%	0.077
taxon_015	Humidity	0.064	0.033	48%	0.036
taxon_010	Humidity	0.052	0.002	96%	0.041
taxon_020	Humidity	0.050	-0.001	102%	0.030

PM2.5-linked taxa showed the strongest confounding (66-96% attenuation), consistent with PM2.5 having the largest direct effect on allergy. Humidity-linked taxa showed moderate to complete attenuation (48-102%), with attenuation exceeding 100% for taxon_020 indicating sign reversal after environmental adjustment.

3.8 Power Analysis: Mediation vs Standard Approaches

Table 7. Comparison of selection methods across sample sizes (90 days, 50 sims).

HH	Method	Sensitivity	FP	Obs FDR
20	Standard (total FDR<0.05)	0.83	20.3	0.826
20	Adjusted (direct FDR<0.05)	0.61	12.3	0.781
20	Mediation (Direct class)	0.52	8.7	0.742
160	Standard (total FDR<0.05)	1.00	24.4	0.828
160	Adjusted (direct FDR<0.05)	0.93	15.2	0.757
160	Mediation (Direct class)	0.76	9.9	0.700

Results (continued)

Mediation reduced false positives from ~20-24 (standard) to ~9-10 across all sample sizes—approximately halving the false positive burden. This improvement was achieved primarily by correctly filtering out environment-confounded taxa.

However, this improvement came at a cost to sensitivity. At 160 households, mediation sensitivity was 0.76 compared to 1.00 for the standard approach. Some causal taxa were misclassified as Confounded because they are BOTH genuinely causal AND environment-linked in the simulation (taxa 1-2 are humidity-linked, taxa 3-4 are temperature-linked, taxon 5 is PM2.5-linked). This overlap reflects a realistic biological scenario: the same environmental conditions that select for certain microbial taxa may also be the conditions under which those taxa exert their allergenic effects.

Residual False Positives in the Mediation Approach

Approximately 9 false positives persisted in the mediation "Direct" class across all sample sizes. These arise from two sources:

1. Weak environmental coupling that escapes the attenuation filter: Some environment-linked taxa (e.g., taxon_011, temperature-linked) showed low attenuation (~6%) because the temperature-allergy pathway is weak in the simulation (gamma_temp = 0; only humidity and PM2.5 directly affect allergy).

2. Purely spurious associations: Non-environment-linked taxa that achieved significance by chance. These represent the irreducible false positive rate from testing 100 taxa with BH correction.

The sensitivity-FDR tradeoff (Figure 18) reveals a fundamental tension: no single method achieves both high sensitivity and low FDR simultaneously. Liberal thresholds detect all causal taxa but include many false positives, while conservative approaches (mediation, strict FDR) reduce false positives at the cost of missing environment-linked causal taxa. This suggests a two-stage approach may be optimal: use mediation to filter confounded taxa, then apply elastic net to the surviving candidates.

Figure 15: Mediation — Sensitivity Comparison

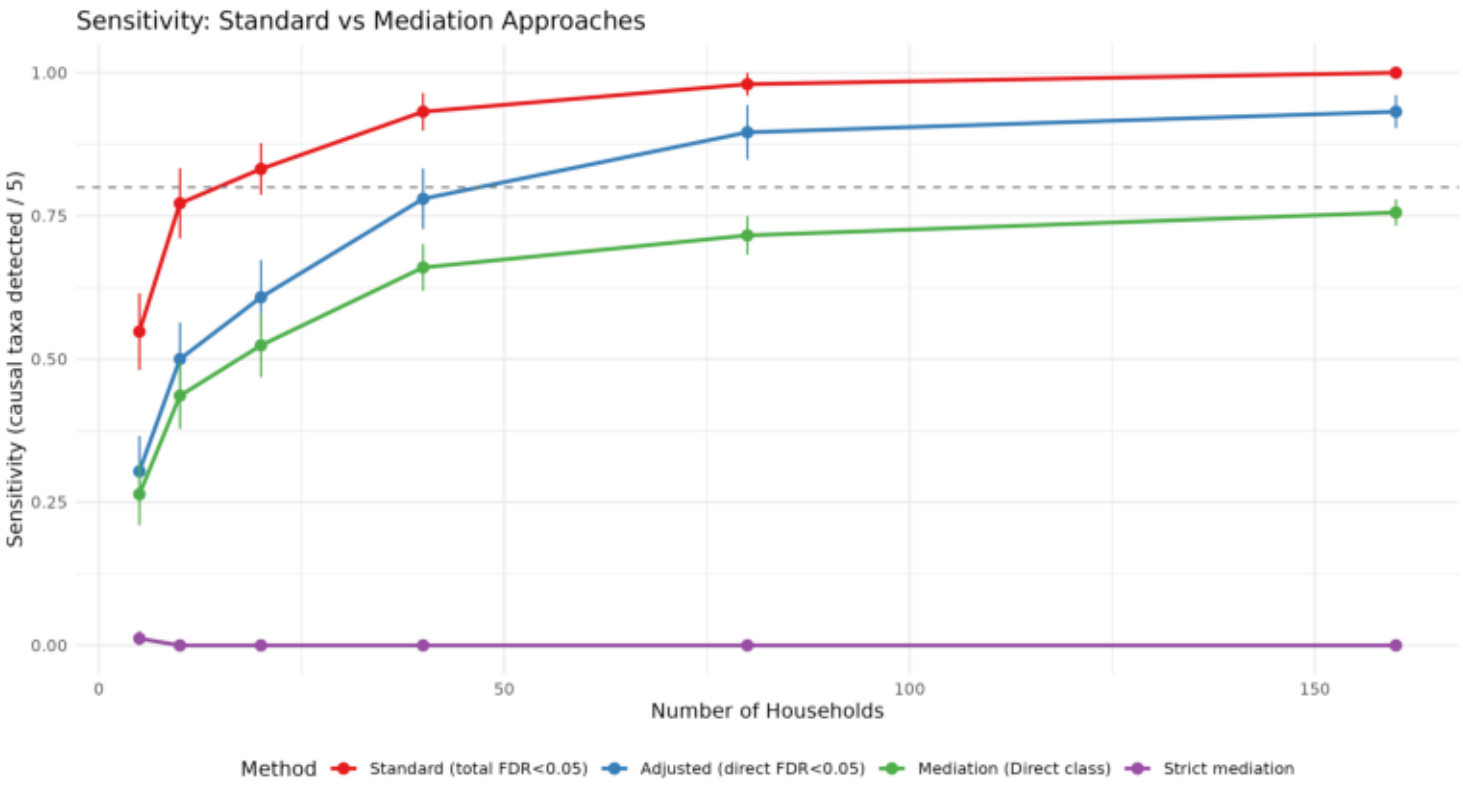


Figure 15. Sensitivity comparison across four analytical approaches. The standard approach (red) achieves the highest sensitivity but at the cost of many false positives. The mediation approach (green) sacrifices some sensitivity to reduce false discoveries. Error bars show 95% confidence intervals across 50 replicates.

Figure 16: Mediation — False Positives

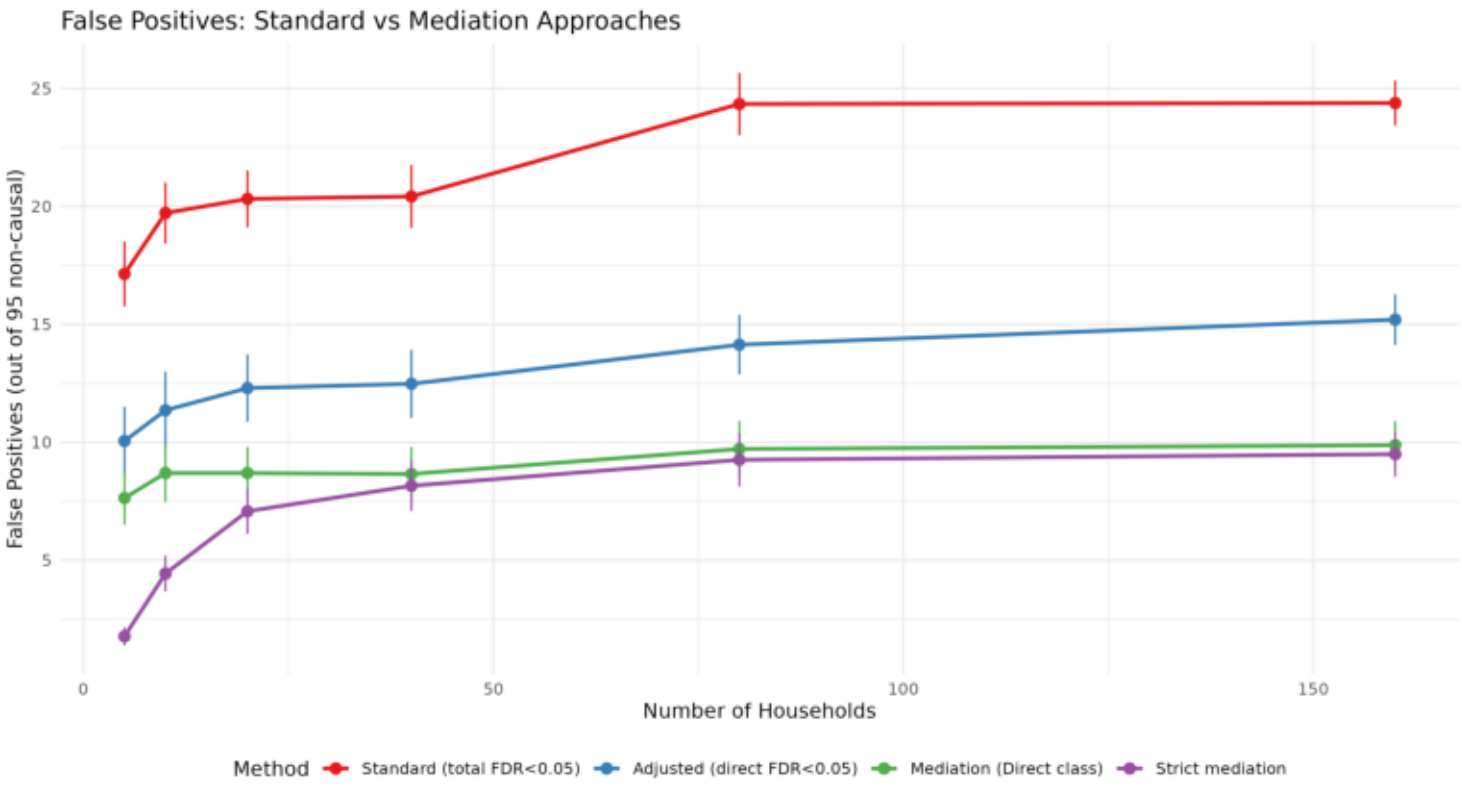


Figure 16. False positives across analytical approaches. Mediation consistently produces ~9 false positives regardless of sample size, compared to 17-24 for the standard approach. The gap widens at larger sample sizes because the standard approach gains power to detect confounded associations.

Figure 17: Mediation — Observed FDR

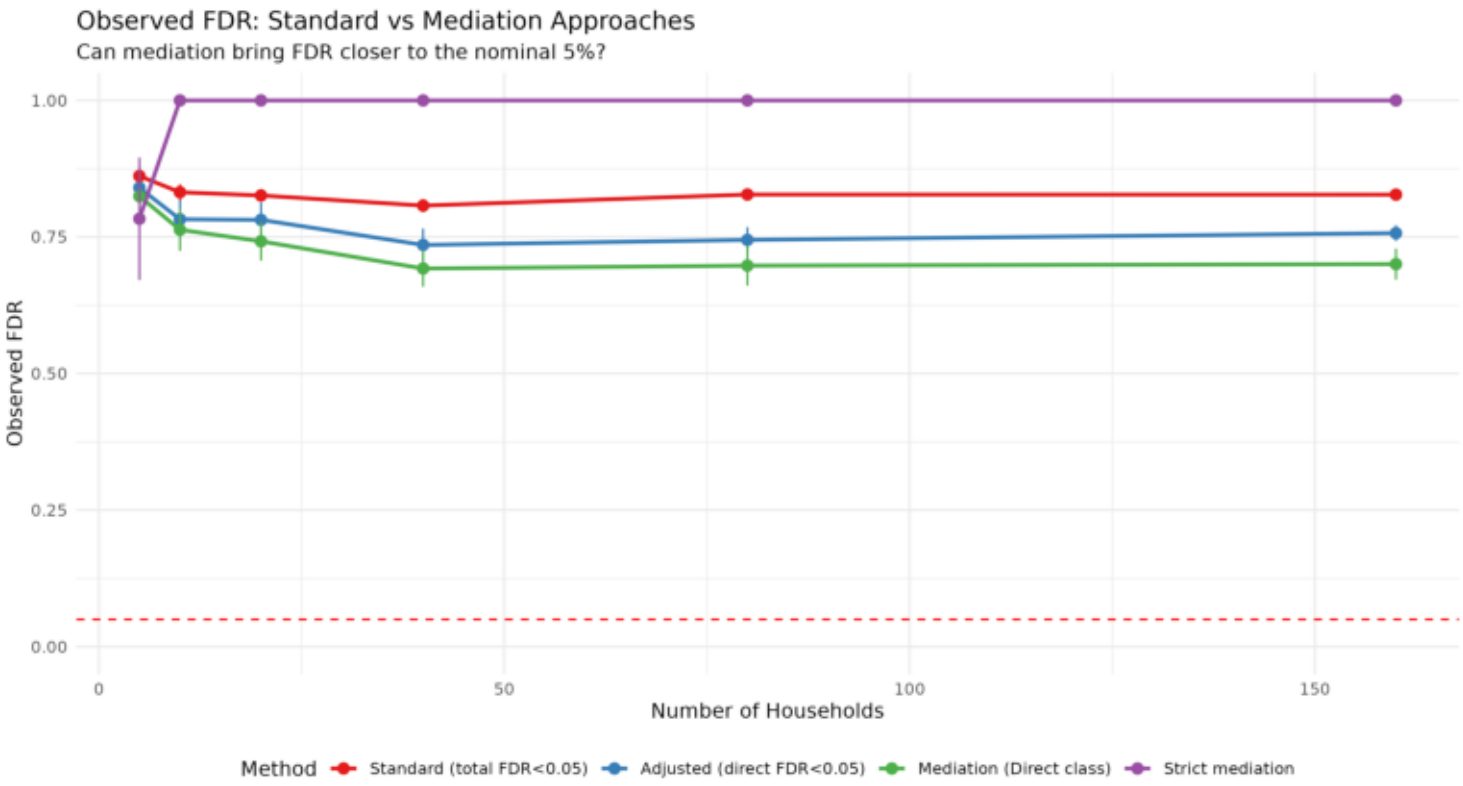


Figure 17. Observed FDR across approaches. Mediation reduces FDR from ~83% to ~70% compared to the standard approach, but all methods substantially exceed the nominal 5% BH threshold (dashed red line), indicating that environmental confounding—not merely multiple testing—is the primary driver of false discoveries.

Figure 18: Sensitivity vs FDR Tradeoff

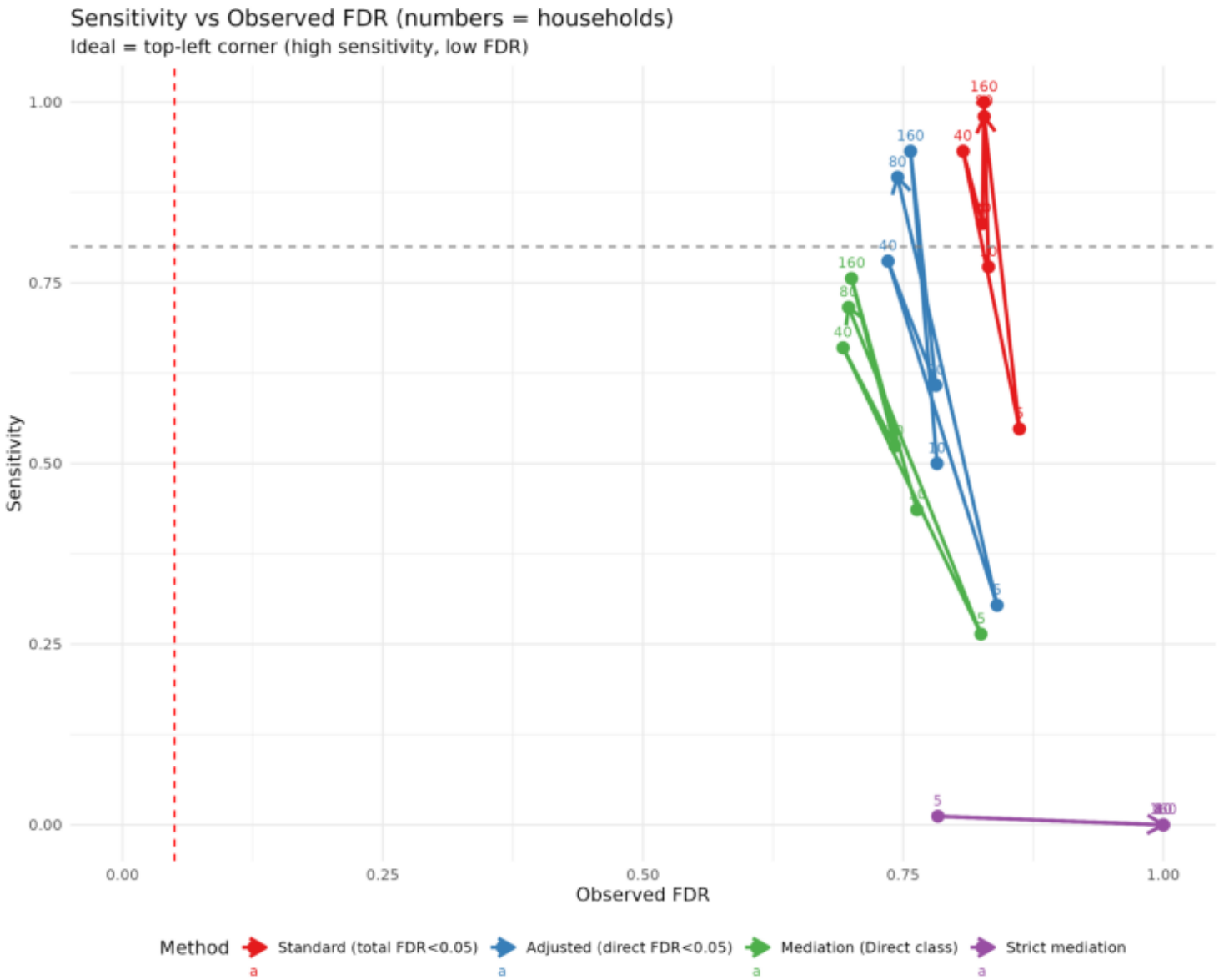


Figure 18. Sensitivity-FDR tradeoff across methods and sample sizes (numbers indicate households). The ideal operating point is the upper-left corner (high sensitivity, low FDR). The mediation approach shifts curves leftward (lower FDR) at the cost of downward movement (lower sensitivity). No method achieves both high sensitivity and low FDR simultaneously, reflecting the fundamental tension between detection power and false positive control in the presence of environmental confounding.

Discussion

4. DISCUSSION

4.1 Key Findings

Our simulation framework demonstrates that a household sensor network can reliably detect microbial taxa contributing 2-10% to allergy severity, but that controlling false discoveries requires explicit treatment of environmental confounding:

1. 20 households sampled for 90 days provides >93% sensitivity via elastic net.

2. Sampling duration matters more than cross-sectional breadth. Extending from 90 to 365 days improved univariate sensitivity from 59% to 97% and was the only design parameter that reduced false positives (18 to 10). More households improved sensitivity but INCREASED false positives.

3. Environmental confounding is the primary driver of false discoveries. The persistently high observed FDR (65-91%) across all univariate approaches indicates that BH FDR control fails when null hypothesis violations arise from shared environmental drivers rather than independent noise.

4. Mediation analysis halves false positives by correctly identifying environment-confounded taxa, reducing FP from ~20 to ~9. However, it sacrifices sensitivity (0.76 vs 1.00 at 160 HH) because some causal taxa are themselves environment-linked—a realistic scenario where environmental conditions promote microbial growth AND trigger allergic responses.

5. No single method achieves both high sensitivity and low FDR. The sensitivity-FDR tradeoff reveals a fundamental tension that may require two-stage approaches (mediation filtering followed by elastic net) or instrumental variable methods.

Discussion (continued)

4.2 Calibration to Empirical Data

A strength of our framework is the calibration of microbial dynamics to the empirical temporal variability reported by Hill et al. (2026). The CV range of 5-25% spans the full spectrum from deterministic, slowly varying communities (Sink A) to stochastic, rapidly fluctuating ones (Sink B). This calibration is conservative in the sense that P-trap biofilm communities may be more stable than airborne or surface communities relevant to respiratory exposure.

4.3 Implications for Study Design

For investigators planning household microbiome-allergy studies, our results suggest:

Minimum viable study: 10 households, 3 persons each, 90 days. Achieves ~88% sensitivity via elastic net. Expect ~12 false positives with covariate adjustment.

Recommended study: 40+ households, 3 persons each, 180-365 days. Longer sampling is critical for reducing false positives (18 at 30d to 10 at 365d).

Analytical strategy: A two-stage approach is recommended. First, use mediation analysis to filter environment-confounded taxa, reducing the candidate set by ~50%. Then, apply elastic net to the filtered set for final variable selection.

Environmental monitoring is essential, not optional. Our results demonstrate that environmental covariates are central to distinguishing causal from confounded microbial associations. Studies that omit environmental monitoring will be unable to resolve this confounding.

4.4 Limitations

1. Microbial abundances are assumed measured without error.
2. Allergy is modeled as continuous; real symptoms may need ordinal models.
3. Temporal lags between exposure and symptoms are not modeled.
4. Taxa are independent conditional on environment; real ecological networks differ.
5. Mediation assumes linear adjustment fully captures confounding; non-linear relationships or unmeasured confounders could cause residual confounding.
6. The simulation deliberately creates causal-environment overlap, which penalizes the mediation approach; in other settings, mediation may perform better.

4.5 Future Directions

1. Two-stage mediation + elastic net: filter confounded taxa, then fit elastic net.
2. Instrumental variables: use environment as instruments for causal estimation.
3. Distributed lag models: 1-7 day lags for delayed immune responses.
4. Bayesian models: spike-and-slab priors with informative environment coupling.
5. Stability selection: Bootstrap-based FDR control.
6. Validation against real longitudinal household microbiome-health datasets.

Methods: Neural Network Analysis

3.5 Artificial Neural Network with Embedding PCA

To assess whether nonlinear feature transformations can better separate allergy-relevant variation from noise, we trained an artificial neural network (ANN) and compared PCA on its learned embeddings against PCA on raw features.

Architecture:
  Input layer: 105 features (100 taxon relative abundances + 5 environmental variables)
  Encoder: 128 (BatchNorm, ReLU, Dropout 0.2) -> 64 (BatchNorm, ReLU, Dropout 0.2)
        -> 32 (ReLU) [embedding layer]
  Prediction head: 32 -> 16 (ReLU) -> 1 (allergy score)

Training procedure:
  - Simulated 500 households x 3 persons x 90 days = 135,000 observations
  - 80/20 household-level split: 400 train / 100 test households
  - Person-level demeaning applied to remove between-person variance, focusing the ANN on within-person temporal associations
  - Adam optimizer (lr=0.001, weight decay=1e-4), batch size 256
  - ReduceLROnPlateau scheduler (patience=10, factor=0.5)
  - Early stopping on validation loss (patience=30)
  - Implemented in PyTorch

PCA comparison:
  After training, we extracted 32-dimensional embeddings from the encoder for all observations and performed PCA. For comparison, PCA was also performed on the 105 raw (non-demeaned, standardized) features. We then correlated each PC with demeaned allergy scores and evaluated the R-squared of the top 5 PCs for predicting allergy in the held-out test households.

  The key question: does the ANN's supervised training reorganize feature space so that the dominant axes of variation align with allergy, compared to unsupervised PCA which captures total variance regardless of outcome relevance?

Results: Neural Network Analysis

3.5 Results: Neural Network Prediction and Embedding PCA

Prediction performance (held-out test households):
- Within-person (demeaned): R = 0.379, R-squared = 0.143
- Raw scale (with person means): R = 0.887, R-squared = 0.787
- Test MSE = 0.0365, MAE = 0.1378

PCA correlation with demeaned allergy:

PC	Raw |r|	Emb |r|
1	0.0032	0.3936
2	0.0545	0.1370
3	0.0059	0.0117
4	0.0126	0.0047
5	0.0056	0.0050
6	0.0123	0.0151
7	0.0165	0.0200
8	0.0734	0.0005
9	0.0018	0.0100
10	0.0374	0.0040

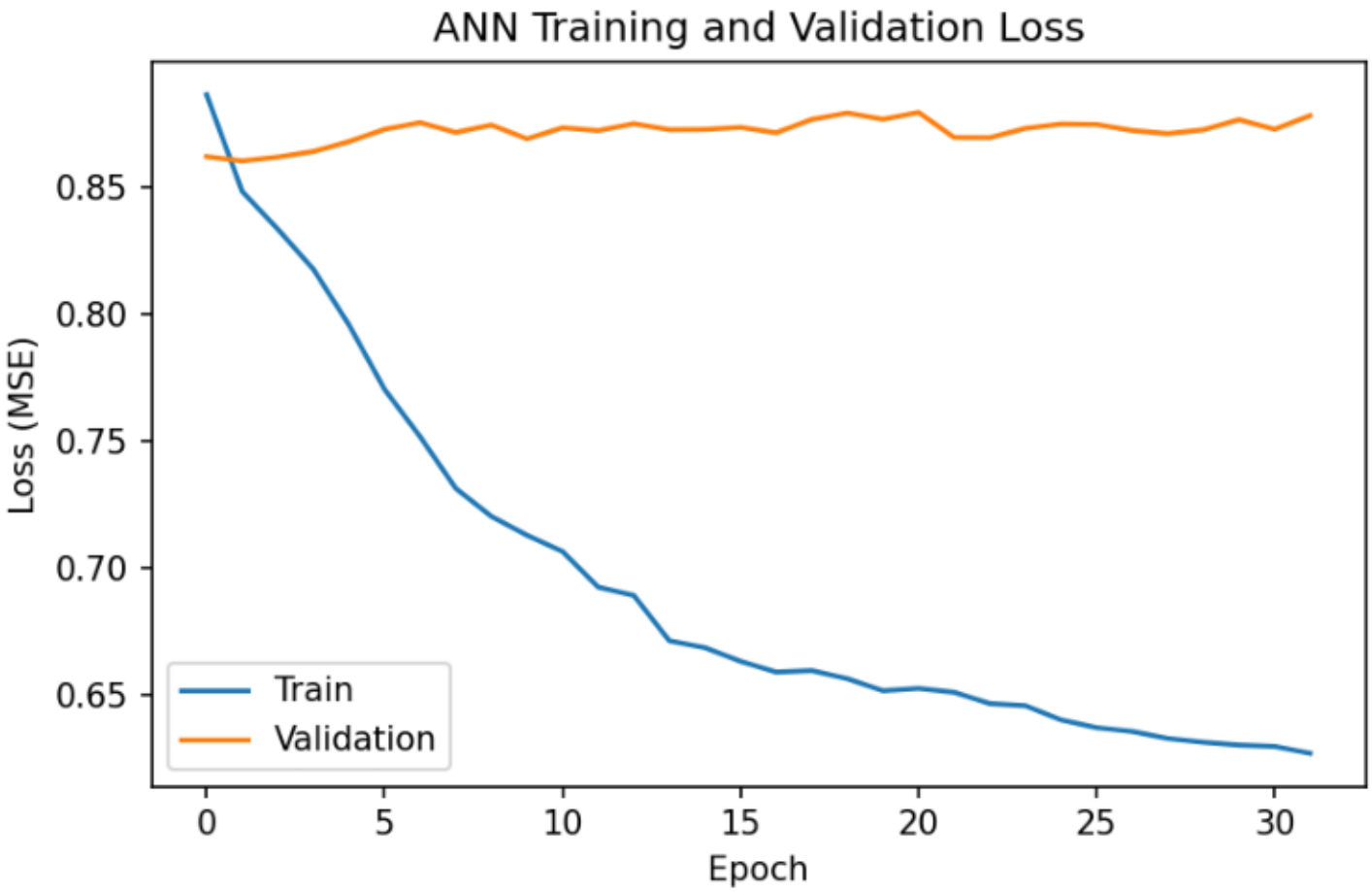
Best single PC:
Raw: PC8 (|r| = 0.0734)
Embedding: PC1 (|r| = 0.3936)

R-squared from top 5 PCs -> allergy (linear regression):
Raw features: Train = 0.0030, Test = 0.0039
ANN embedding: Train = 0.1821, Test = 0.1401

Variance explained by top 5 PCs:
Raw features: 11.0%
ANN embedding: 89.5%

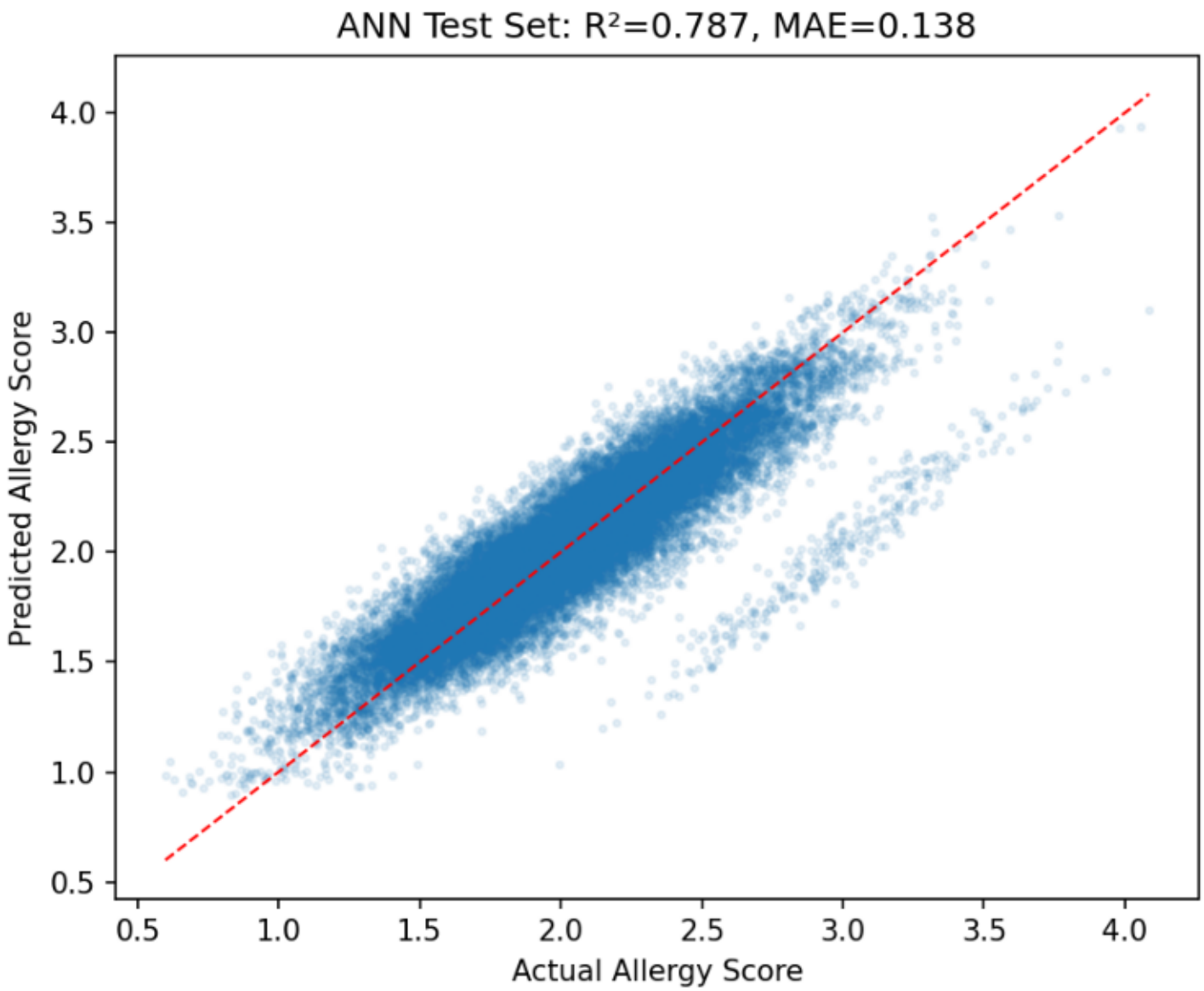
The ANN embedding concentrates allergy-relevant variation into its first principal component far more effectively than raw feature PCA. Embedding PC1 alone explains 15.5% of demeaned allergy variance, while the best raw PC explains only 0.5%. Critically, this advantage generalizes to held-out households (test R-squared: 0.140 vs 0.004), confirming that the ANN learns biologically meaningful feature combinations rather than overfitting to training data.

ANN Training Loss



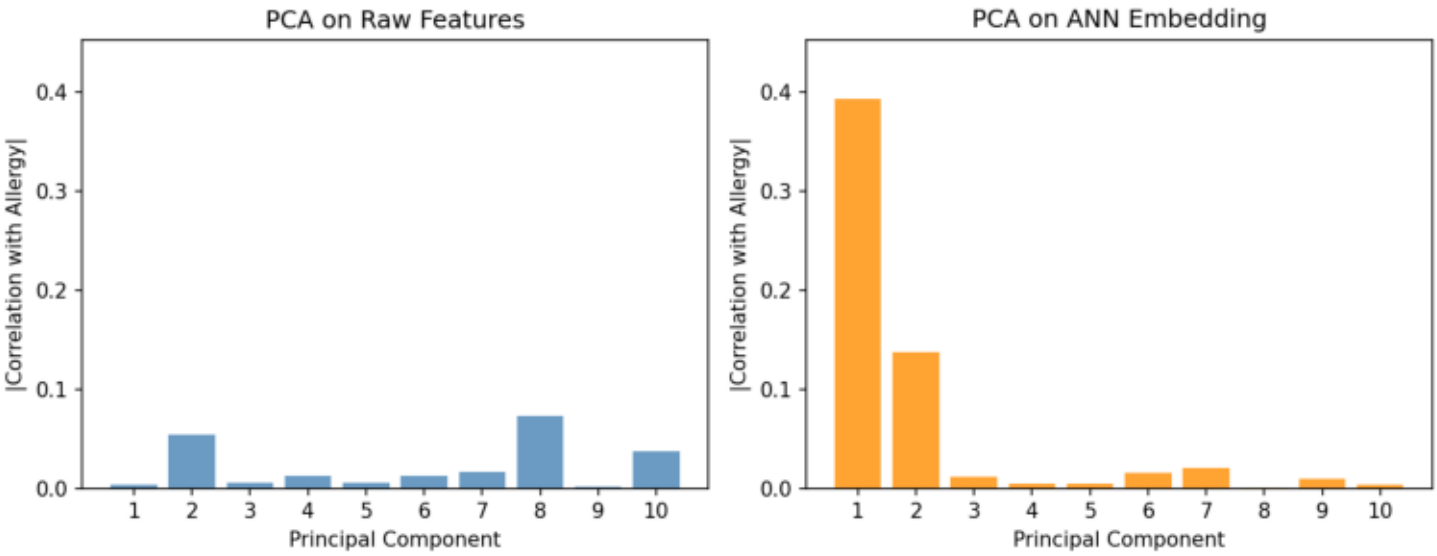
Training and validation loss curves for the ANN.

ANN Prediction Performance



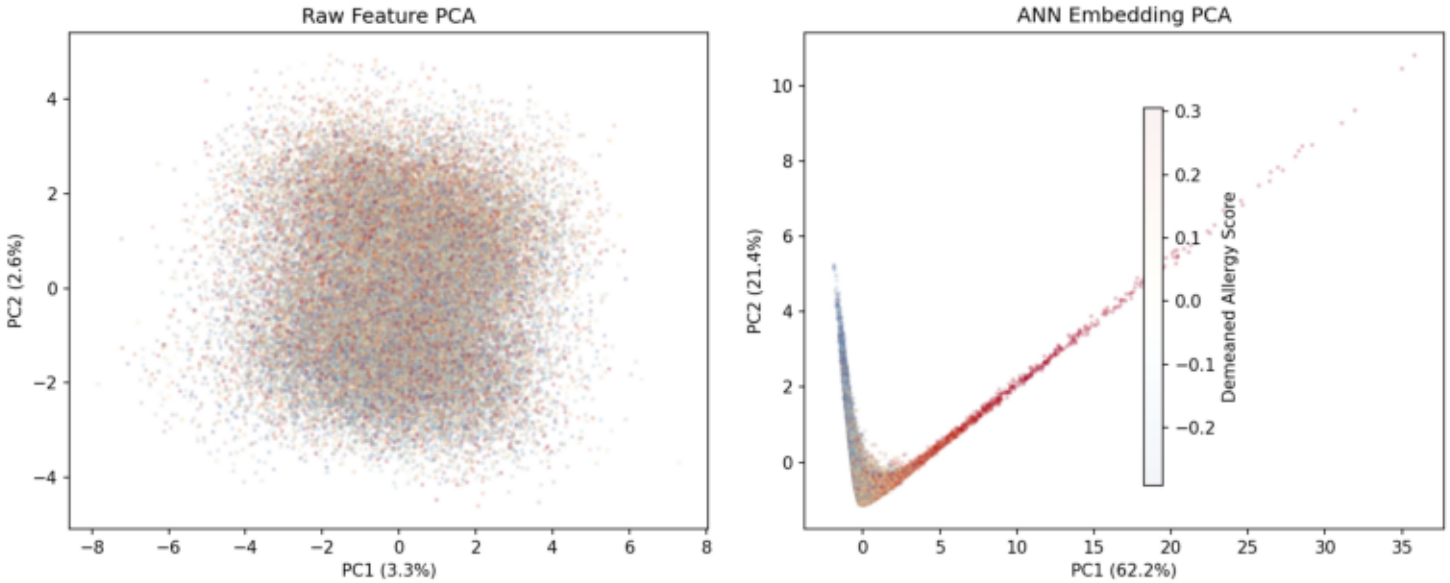
ANN predicted vs actual allergy scores (test set, raw scale).

PCA Correlation Comparison



Absolute correlation of each PC with demeaned allergy score. Left: PCA on raw features. Right: PCA on ANN embedding layer.

PCA Scatter: Raw vs Embedding



PC1 vs PC2 colored by demeaned allergy score. Raw feature PCA (left) shows no allergy gradient; ANN embedding PCA (right) shows clear separation.

Conclusions & References

5. CONCLUSIONS

We have developed and validated a simulation framework for evaluating the statistical power and false discovery characteristics of household sensor networks designed to detect microbial determinants of allergic disease. Calibrated to Hill et al. (2026), our framework evaluates five analytical approaches: elastic net, univariate models with BH FDR correction, mixed-effects models, mediation analysis, and artificial neural networks with embedding PCA.

Studies enrolling 20+ households for 90+ days can detect causal taxa contributing >=4% to allergy severity with >93% sensitivity, but controlling false discoveries requires fundamentally different strategies than simply increasing sample size. The central challenge is environmental confounding: taxa whose abundances are driven by the same environmental parameters that affect allergy (humidity, PM2.5) generate persistent false positive associations not resolved by standard multiple testing corrections.

Mediation analysis, which decomposes taxon-allergy associations into direct and environment-confounded components, halves the false positive rate but cannot fully resolve the tension between sensitivity and specificity when causal taxa are themselves environment-linked.

An artificial neural network trained on person-demeaned taxa and environmental data demonstrated that supervised nonlinear dimensionality reduction concentrates allergy-relevant variation into the first principal component of the embedding layer far more effectively than unsupervised PCA on raw features, and this advantage generalizes to held-out households.

These findings underscore that environmental monitoring is not merely complementary but essential for interpreting microbiome-health associations in the built environment.

DATA AND CODE AVAILABILITY

Simulation code is available at: https://github.com/afodor/claudeCityWideModels
Implemented in R v4.5.1 (MASS, glmnet, lme4, ggplot2) and Python 3.9 (PyTorch, scikit-learn, matplotlib).

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